NUCLEAR MAGNETIC RESONANCE SPECTROSCOPY

BY

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3. Introduction

Nuclear magnetic resonance (NMR) spectroscopy allows us not only to detect atomic nuclei but also tells us about their environment, i.e, which type of atoms they are attached with. For example the hydrogen of propanol's hydroxyl group is different from the hydrogens of its carbon skeleton—it can be displaced by sodium metal, for example.

NMR (actually ^{1}H, or proton, NMR) can easily distinguish between these two sorts of hydrogens. Moreover, it can also distinguish between all the other different sorts of hydrogen atoms present. Likewise, carbon (or rather ^{13}C) NMR can easily distinguish between the three different carbon atoms.

3.1 NUCLEAR SPIN STATES

Many atomic nuclei have a property called spin: **the nuclei behave as if they were spinning**. In fact, any atomic nucleus that possesses either odd mass, odd atomic number, or both has a quantized spin angular momentum and a magnetic moment. The more common nuclei that possess spin include $^{1}{}_{1}H$, $^{2}{}_{1}H$, $^{13}{}_{6}C$, $^{14}{}_{7}N$, $^{17}{}_{8}O$, and $^{19}{}_{9}F$. Notice that the nuclei of the ordinary (most abundant) isotopes of carbon and oxygen, $^{12}{}_{6}C$ and $^{16}{}_{8}O$, are not included among those with the spin property. However, the nucleus of the ordinary hydrogen atom, the proton, does have spin.

For each nucleus with spin, the number of allowed spin states it may adopt is quantized and is determined by its nuclear spin quantum number I. For each nucleus, the number I is a physical constant, and there are 2I+1 allowed spin states with integral differences ranging from +I to −I.

For instance, a proton (hydrogen nucleus) has the spin quantum number I= 1/2 and has two allowed spin states [2(1/2) + 1 =2] for its nucleus: −1/2 and +1/2 . For the chlorine nucleus, I= 3/2 and there are four allowed spin states [2(3/2) +1 =4]:− 3/2 ,− 1/ 2 ,+ 1/ 2 ,and + 3/ 2 . Table below gives the spin quantum numbers of several nuclei.

Table 1.1 Spin number of I of some nuclei

Mass number	Atomic number	Spin number I	Examples
Odd	Odd or even	$\frac{1}{2}$	^{1}H, ^{13}C, ^{15}N, ^{19}F, ^{31}P
		$\frac{3}{2}$	^{11}B, ^{35}Cl, ^{37}Cl, ^{79}Br, ^{81}Br
		$\frac{5}{2}$	^{127}I, ^{17}O
Even	Even	Zero (no spin)	^{12}C, ^{16}O, ^{32}S, ^{34}S
Even	Odd	1	^{14}N, ^{2}H (or D)
		3	^{10}B

3.2 NUCLEAR MAGNETIC MOMENTS

In the absence of an applied magnetic field, all the spin states of a given nucleus are of equivalent energy (degenerate), and in a collection of atoms, all of the spin states should be almost equally populated, with the same number of atoms having each of the allowed spins.

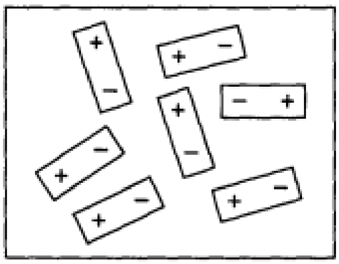


Fig. 1.2. Orientation of nuclear magnetic dipoles in the absence of an external magnetic field.

However, Spin states are not of equivalent energy in an applied magnetic field because the nucleus is a charged particle, and any moving charge generates a magnetic field of its own. Thus, the nucleus has a magnetic moment μ (meu) generated by its charge and spin. A hydrogen nucleus may have a clockwise (+1/ 2) or counterclockwise (– 1/2) spin, and the nuclear magnetic moments (μ) in the two cases are pointed in opposite directions. In an applied magnetic field, all protons have their magnetic moments either aligned with the field or opposed to it. Figure 1.3 illustrates these two situations. Hydrogen nuclei can adopt only one or the other of these orientations with respect to the applied field. The spin state +1/2 is of lower energy since it is aligned with the field, while the spin state −1/2 is of higher energy since it is opposed to the applied field.

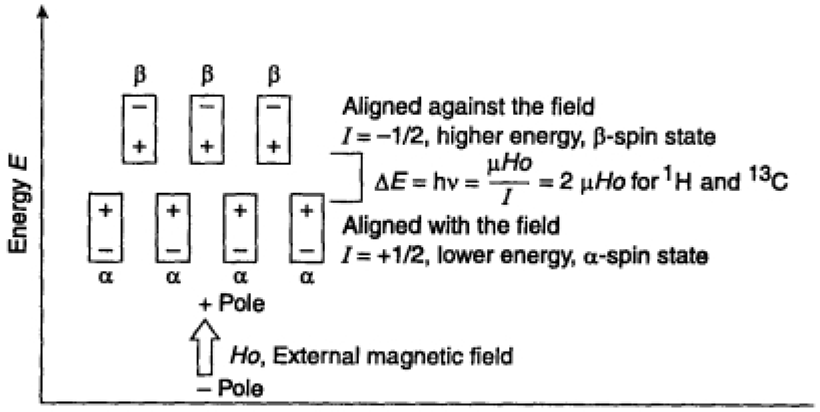


Fig. 1.3. Orientation of nuclear magnetic dipoles in an external magnetic field H_0.

We can consider these nuclei as the two magnets. The aligned configuration of magnets is stable (low energy). However, the opposed (not aligned) configuration is unstable (high energy). When an external magnetic field is applied, the degenerate spin states split into two states of unequal energy, as shown in Figure 1.3.

3.3 ABSORPTION OF ENERGY (Spin flipping)

The **nuclear magnetic resonance phenomenon occurs** when nuclei aligned with an applied field are **induced to absorb energy** and change their spin orientation with respect to the applied field. It is called **spin flipping**. Figure 1.4 illustrates this process for a hydrogen nucleus.

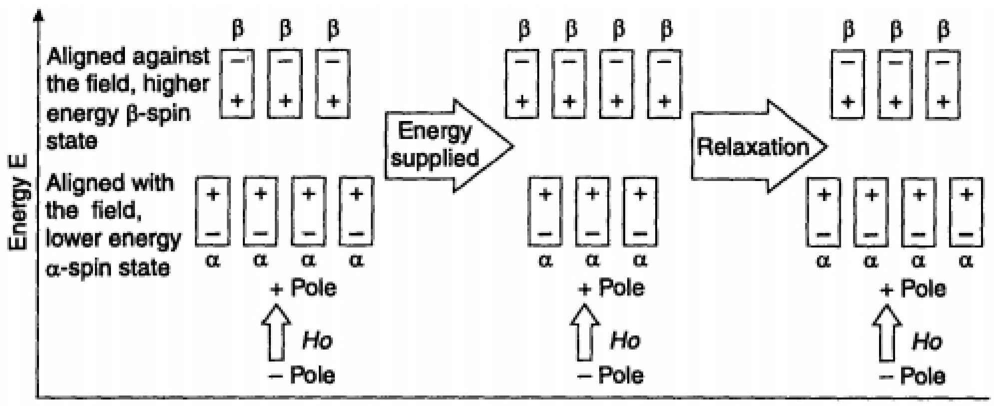


Fig. 1.4. Absorption of energy (ΔE) flips a nuclear spin, converting the lower energy orientation into the higher one.

The energy absorption is a quantized process, and the energy absorbed must equal the energy difference between the two states involved.

$E_{absorbed} = [E(-1/2) - E(+1/2)] = h\upsilon$

In practice, this energy difference is a function of the strength of the applied magnetic field B_0, as illustrated in Figure 3.6.

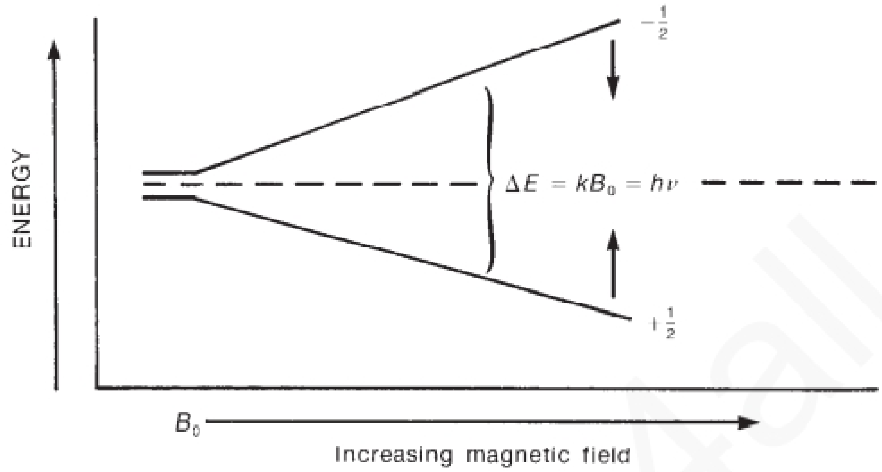


FIGURE 3.6 The spin-state energy separation as a function of the strength of the applied magnetic field B_0.

The stronger the applied magnetic field, the greater the energy difference between the possible spin states:

$$\Delta E = f(B_0) \qquad \textbf{Equation 3.3}$$

The magnitude of the energy-level separation also depends on the particular nucleus involved. Each nucleus (hydrogen, chlorine, and so on) has a different **ratio of magnetic moment to angular momentum** since each has different charge and mass. This ratio, called the **magnetogyric ratio γ**, is a constant for each nucleus and determines the energy dependence on the magnetic field:

$$\Delta E = f(\gamma B_0) = h\nu \qquad \textbf{Equation 3.4}$$

Since the angular momentum of the nucleus is quantized in units of h/2π, the final equation takes the form

$$\Delta E = \gamma\left(\frac{h}{2\pi}\right) B_0 = h\nu \qquad \textbf{Equation 3.5}$$

Solving for the frequency of the absorbed energy,

$$\upsilon = \left(\frac{\gamma}{2\pi}\right) B_0 \qquad \textbf{Equation 3.6}$$

If the correct value of γ for the proton is substituted, one finds that an unshielded proton should absorb radiation of frequency **42.6 MHz** in a field of strength **1 Tesla (10,000 Gauss**) or radiation of frequency **60.0 MHz** in a field of strength **1.41 Tesla (14,100 Gauss).** Table 3.2 shows the field strengths and frequencies at which several nuclei have resonance (i.e., absorb energy and make spin transitions).

TABLE 3.2
FREQUENCIES AND FIELD STRENGTHS AT WHICH SELECTED NUCLEI HAVE THEIR NUCLEAR RESONANCES

Isotope	Natural Abundance (%)	Field Strength, B_0 (Tesla[a])	Frequency, ν (MHz)	Magnetogyric Ratio, γ (radians/Tesla)
^{1}H	99.98	1.00	42.6	267.53
		1.41	60.0	
		2.35	100.0	
		4.70	200.0	
		7.05	300.0	
^{2}H	0.0156	1.00	6.5	41.1
^{13}C	1.108	1.00	10.7	67.28
		1.41	15.1	
		2.35	25.0	
		4.70	50.0	
		7.05	75.0	
^{19}F	100.0	1.00	40.0	251.7
^{31}P	100.0	1.00	17.2	108.3

[a] 1 Tesla = 10,000 Gauss.

3.4.1 The Mechanism of Absorption (Resonance)

Although many nuclei are capable of exhibiting magnetic resonance, the organic chemist is mainly interested in hydrogen and carbon resonances. For a proton (the nucleus of a hydrogen atom), if the applied magnetic field has a strength of approximately **1.41 Tesla**, the difference in energy between the two spin states of the proton is about **2.39 × 10^{-5} kJ/mole**. Radiation with a frequency of about **60 MHz (60,000,000 Hz),** which lies in the radiofrequency (RF) region of the electromagnetic spectrum, corresponds to this energy difference. Other nuclei have both larger and smaller energy differences between spin states than do hydrogen nuclei.

The earliest nuclear magnetic resonance spectrometers applied a **variable magnetic field with a range of strengths near 1.41 Tesla and supplied a constant radiofrequency radiation of 60 MHz.** They effectively induced transitions only among proton (hydrogen) spin states in a molecule and were **not useful for other nuclei**. Separate instruments were required to observe transitions in the nuclei of other elements, such as carbon and phosphorus. Fourier transform instruments, which are in common use today, are equipped to observe the nuclei of several different elements in a single instrument. Instruments operating at frequencies of **300 and 400 MHz** are now quite common, and instruments with frequencies above **600 MHz** are found in the larger research universities.

3.4.2 THE MECHANISM OF ABSORPTION (RESONANCE)

Protons absorb energy due to their precessional frequency. Just like spinning top, **spinning protons began to precess under the influence of magnetic field** just as spinning top began to wobble under the influence of gravitational field.

When the magnetic field is applied, the nucleus begins to precess about its own axis of spin with angular frequency, which is sometimes called its **Larmor frequency**.

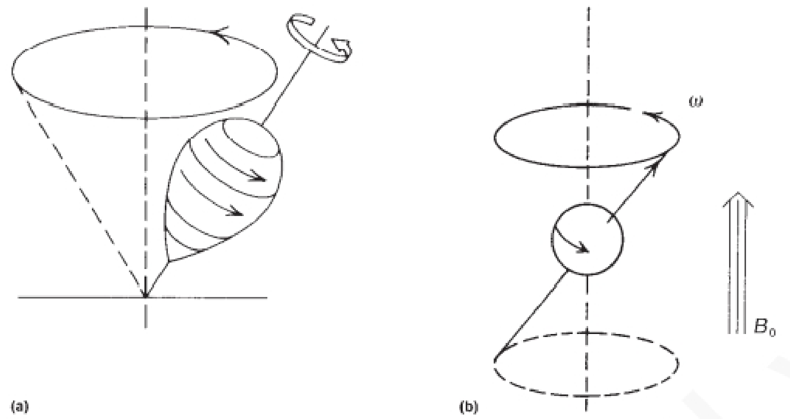


FIGURE 3.7 (a) A top precessing in the earth's gravitational field; (b) the precession of a spinning nucleus resulting from the influence of an applied magnetic field.

This frequency at which a proton precesses is directly proportional to the strength of the applied magnetic field; **the stronger the applied field, the higher the rate (angular frequency ω) of precession**. For a proton, if the applied field is 1.41 Tesla (14,100 Gauss), the frequency of precession is approximately 60 MHz.

Since the nucleus has a charge, the precession generates an oscillating electric field of the same frequency. **If radiofrequency waves of this frequency are supplied to the precessing proton, the energy can be absorbed.** That is,

when the frequency of the oscillating electric field component of the incoming radiation just matches the frequency of the electric field generated by the precessing nucleus, the two fields can couple, and energy can be transferred from the incoming radiation to the nucleus, thus causing a spin change.

This condition is called **resonance**, and the nucleus is said to have resonance with the incoming electromagnetic wave. Figure 3.8 schematically illustrates the resonance process.

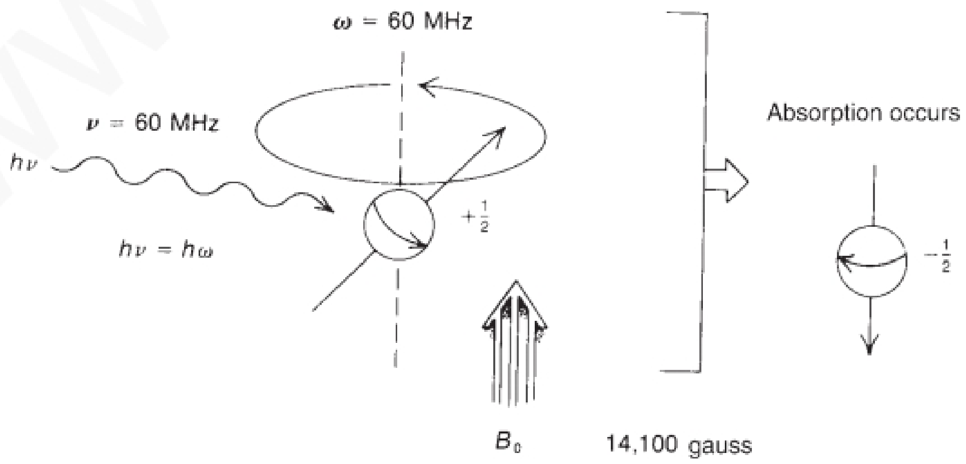


FIGURE 3.8 The nuclear magnetic resonance process; absorption occurs when $\upsilon = \omega$.

3.5 POPULATION DENSITIES OF NUCLEAR SPIN STATES

At applied magnetic field of 1.41 Tesla, i.e., at 60 MHz, the energy difference between the alpha and beta spin states of proton is about 2.39×10^{-5} kJ/mole. Due to this very small energy difference, **there is slight excess of nuclei in the lower-energy (alpha) spin state.** The magnitude of this difference can be calculated using the **Boltzmann distribution equations.**

Using Boltzmann Equation, one can calculate that at 298 K (25°C), for an instrument operating at 60 MHz there are 1,000,009 nuclei in the lower (favored) spin state for every 1,000,000 that occupy the upper spin state:

In other words, in approximately 2 million nuclei, there are only 9 more nuclei in the lower spin state. Let us call this number (9) the excess population.

The excess nuclei are the ones that allow us to observe resonance.

If we increase the operating frequency of the NMR instrument, the energy difference between the two states increases, which causes an increase in this excess. It means by increasing the applied magnetic field the excess population can be increased.

3.6 THE CHEMICAL SHIFT AND SHIELDING

3.6.1 Different protons resonate at different frequencies:

Nuclear magnetic resonance has great utility because not all protons in a molecule have resonance at exactly the same frequency. This variability is due to the fact that the protons in a molecule are surrounded by electrons and exist in slightly different electronic (magnetic) environments from one another. The valence-shell electron densities vary from one proton to another.

3.6.2 Local diamagnetic shielding or diamagnetic anisotropy:

The protons are shielded by the electrons that surround them. In an applied magnetic field, the valence electrons of the protons are caused to circulate. This circulation, called a **local diamagnetic current**, generates a counter magnetic field that opposes the applied magnetic field.

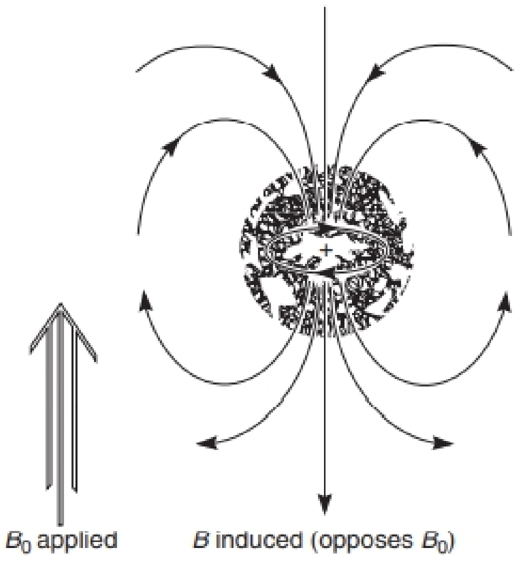


FIGURE 3.10 Diamagnetic anisotropy—the diamagnetic shielding of a nucleus caused by the circulation of valence electrons.

Figure 3.10 illustrates this effect, which is called **diamagnetic shielding or diamagnetic anisotropy**. Just like the flow of a current through a wire induces a magnetic field, **circulation of electrons around a nucleus, in an atom, the local diamagnetic current generates a secondary, induced magnetic field that has a direction opposite that of the applied magnetic field.**

This diamagnetic anisotropy shields the nearby proton from applied magnetic field. **Strength of this shield is directly proportional to the electron density around the proton**. The greater the electron density around a nucleus, the greater the induced counter field that opposes the applied field. The difference between the energy of the nucleus lying with the applied field and against the applied field decreases and so it absorbs the radio frequency at low value and precesses at a lower frequency. **This means that it also absorbs radiofrequency radiation at this lower frequency.**

Each proton in a molecule is in a slightly different chemical environment and consequently has a slightly different amount of electronic shielding, which results in a slightly different resonance frequency.

3.6.3 Chemical Shift Internal reference compound

The **differences in resonance frequency of different protons are very small**. For instance, the difference between the resonance frequencies of the protons in chloromethane and those in fluoromethane is only 72 Hz when the applied field is 1.41 Tesla.

Since the radiation used to induce proton spin transitions at that magnetic field strength is of a frequency near 60 MHz, the difference between chloromethane and fluoromethane represents a change in frequency of only slightly more than one part per million!

It is very difficult to measure exact frequencies to that precision; hence, **no attempt is made to measure the exact resonance frequency of any proton**.

Instead, a **reference compound** is placed in the solution of the substance to be measured, and the **resonance frequency of each proton in the sample is measured relative to the resonance frequency of the protons of the reference substance.** In other words, the frequency difference is measured directly.

The standard reference substance that is used universally is **tetramethylsilane, $(CH_3)_4$ Si, also called TMS**. This compound was chosen initially because the protons of its methyl groups are more shielded than those of most other known compounds. At that time, no compounds that had better-shielded hydrogens than TMS were known, and it was assumed that TMS would be a **good reference substance since it would mark one end of the range**. Thus, when another compound is measured, the resonances of its protons are reported in terms of how far (in Hertz) they are shifted from those of TMS.

The shift from TMS for a given proton depends on the strength of the applied magnetic field. In an applied field of 1.41 Tesla the resonance of a proton is approximately 60 MHz, whereas in an applied field of 2.35 Tesla (23,500 Gauss) the resonance appears at approximately 100 MHz.

The ratio of the resonance frequencies is the same as the ratio of the two field strengths:

Hence, for a given proton, the shift (in Hertz) from TMS is 5/3 larger in the 100-MHz range (B_0 =2.35 Tesla) than in the 60-MHz range (B_0 = 1.41 Tesla).

This can be confusing for workers trying to compare data if they have spectrometers that differ in the strength of the applied magnetic field. The confusion is easily overcome if one defines a new parameter that is independent of field strength— for instance,

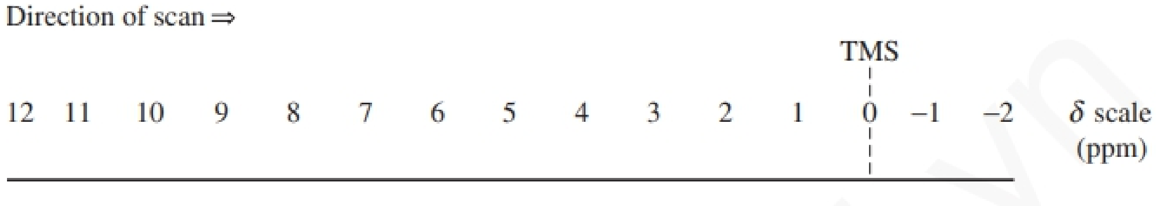
by dividing the shift in Hertz of a given proton by the frequency in megahertz of the spectrometer with which the shift value was obtained. In this manner, a field-independent measure called the chemical shift (d) is obtained

Delta (δ) = $\frac{Distance\ in\ Hz\ from\ TMS}{spectrometer\ frequency\ in\ MHz}$

The chemical shift in δ units expresses the amount by which a proton resonance is shifted from TMS, in parts per million (ppm), of the spectrometer's basic operating frequency. Values of **δ** for a given proton are always the same irrespective of whether the measurement was made at 60 MHz (B_0 = 1.41 Tesla) or at 100 MHz (B_0 = 2.35 Tesla). For instance, at 60 MHz the shift of the protons in CH_3Br is 162 Hz from TMS, while at 100 MHz the shift is 270 Hz. However, both of these correspond to the same value of **δ** (2.70 ppm):

Delta (δ) = $\frac{162\ Hz}{60\ MHz}$z = $\frac{270\ Hz}{100\ MHz}$ = 2.70 ppm

By agreement, most workers report chemical shifts in delta (δ) units, or parts per million (ppm), of the main spectrometer frequency. On this scale, the resonance of the protons in TMS comes at exactly 0.00 ppm (by definition). The NMR spectrometer actually scans from high values to low ones (as will be discussed further). Following is a typical chemical shift scale with the sequence of values that would be found on a typical NMR spectrum chart.



3.7 THE NUCLEAR MAGNETIC RESONANCE SPECTROMETER

3.7.1 The Continuous-Wave (CW) Instrument

Figure 3.11 schematically illustrates the basic elements of a classical 60-MHz NMR spectrometer.

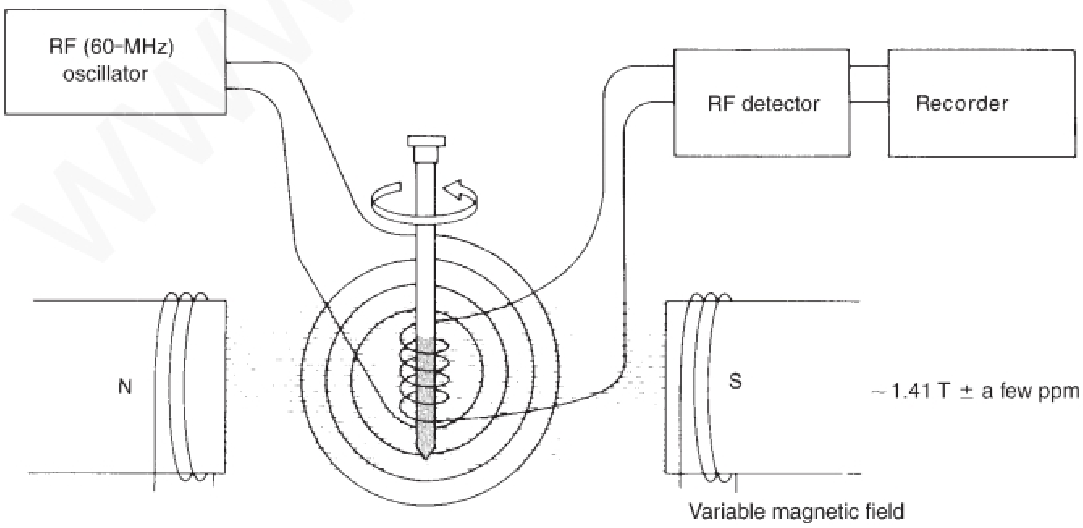


FIGURE 3.11 The basic elements of the classical nuclear magnetic resonance spectrometer.

1-The sample is dissolved in a solvent containing no interfering protons (usually CCl4 or CDCl3), and a small amount of TMS is added to serve as an internal reference.
2-The sample cell is a small cylindrical glass tube that is suspended in the gap between the faces of the pole pieces of the magnet. The sample is spun around its axis to ensure that all parts of the solution experience a relatively uniform magnetic field.
3- In the magnet gap is a coil attached to a 60-MHz radiofrequency (RF) generator. This coil supplies the radio frequency to change the spin orientations of the protons.
4-Perpendicular to the RF oscillator coil is a detector coil.
5-When **no absorption of energy** is taking place, the detector **coil picks up none of the energy** given off by the RF oscillator coil. When the **sample absorbs energy,** however, the **reorientation of the nuclear spins induces a RF signal** in the plane of the detector coil, and the instrument responds by recording this as a resonance signal, or peak.
6-At a constant field strength, the different types of protons in a molecule **precess** at slightly different frequencies.

7- The CW NMR spectrometer uses a **constant-frequency RF signal** and **varies the magnetic field strength**. As the magnetic field strength is increased, the precessional frequencies of all the protons increase. **When the precessional frequency of a given type of proton reaches 60 MHz, it has resonance**.

8- Changing the field in this way systematically brings all of the different types of protons in the sample into resonance. As the field strength is increased linearly, a pen travels across a recording chart. A typical spectrum is recorded as in Figure 3.12.

9-As the pen travels from left to right, the magnetic field is increasing. As each chemically distinct type of proton comes into resonance, it is recorded as a peak on the chart. The **peak at d = 0 ppm is due to the internal reference compound TMS.** Since highly shielded protons precess more slowly than relatively unshielded protons, it is necessary to increase the field to induce them to precess at 60 MHz. Hence, **highly shielded protons appear to the right of this chart, and less shielded, or deshielded, protons appear to the left.** The region of the chart to the left is sometimes said to be **downfield** (or at low field), and that to the right, **upfield** (or at high field).

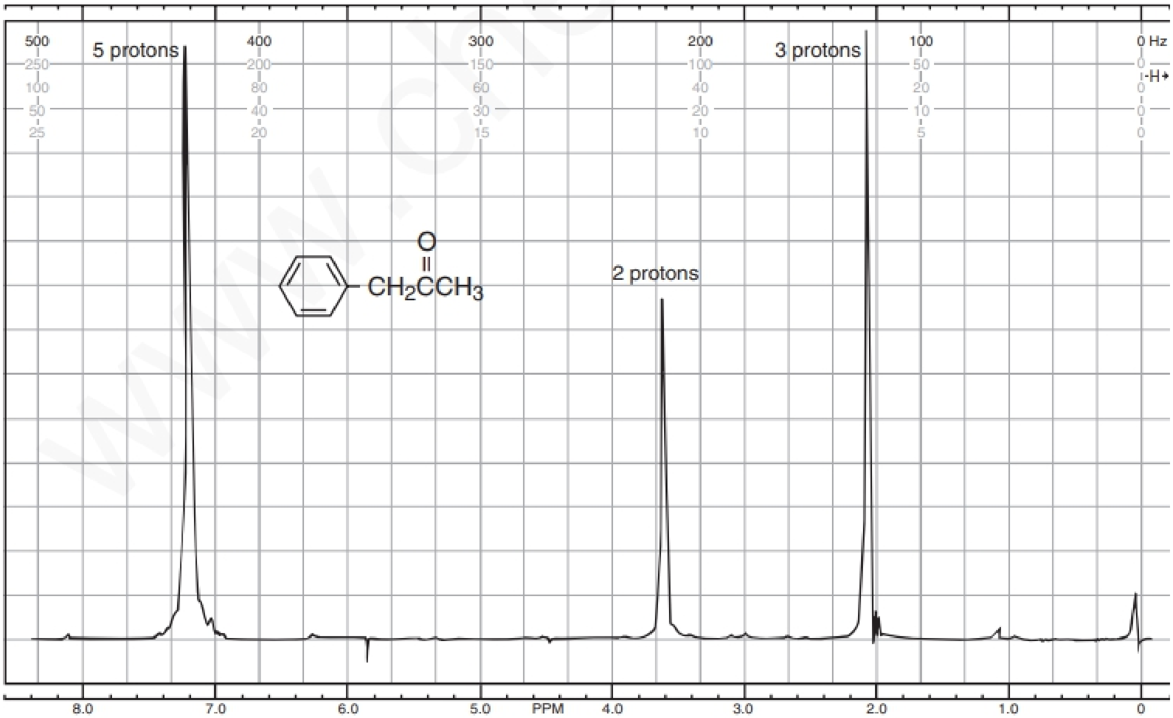


FIGURE 3.12 The 60-MHz ^{1}H nuclear magnetic resonance spectrum of phenylacetone (the absorption

10- Instruments that vary the magnetic field in a continuous fashion, scanning from the downfield end (left)to the up field end (right)of the spectrum, are called continuous-wave (CW) instruments. Because the chemical shifts of the peaks in this spectrum are calculated from frequency differences from TMS, **this type of spectrum (Fig. 3.12) is said to be a frequency-domain spectrum.**

3.7.2 The Pulsed Fourier Transform (FT) Instrument:

The CW type of NMR spectrometer, operates by exciting the every nuclei of one type at a time.

1-An alternative approach, common to modern, sophisticated instruments, is to use a powerful but short burst of energy, called a **pulse**, that excites all of the magnetic nuclei in the molecule simultaneously.

2-The source is turned on and off very quickly, generating a pulse that actually contains a range of frequencies. This pulse is great enough to excite all of the distinct types of hydrogens in the molecule at once with this single burst of energy.

3-When the pulse is discontinued, the excited nuclei begin to lose their excitation energy and return to their original spin state, or relax. As each excited nucleus relaxes, it emits electromagnetic radiation.

4-Since the molecule contains many different nuclei, many different frequencies of electromagnetic radiation are emitted simultaneously. This emission is called a **free-induction decay (FID) signal**.

5-The FID is a superimposed combination of all the frequencies emitted and can be quite complex. This is usually called a **time domain signal**.

6-We usually apply a mathematical method called a Fourier transform (FT) on it. In this way we can convert the time-domain signal to a frequency-domain signal, which is the standard format for a spectrum obtained by a CW instrument.

3.7.3 Differences between two methods:

1- CW spectrum; a pulsed experiment is much faster, and a measurement of an FID can be performed in a few seconds.

2- With a computer and fast measurement, it is possible to repeat and average the measurement of the FID signal. This is a real advantage when the sample is small, in which case the FID is weak in intensity and has a great amount of noise associated with it. **Noise is random electronic signals** that are usually visible as fluctuations of the baseline in the signal (Fig. 3.17). Since noise is random, its intensity does not increase as many iterations of the spectrum are added together. Using this procedure, one can show that the signal-to-noise ratio improves.

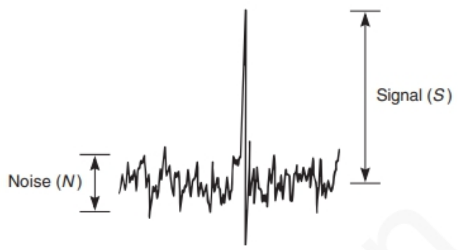


FIGURE 3.17 The signal-to-noise ratio.

3- Pulsed FT-NMR is therefore especially suitable for the examination of nuclei that are not very abundant in nature, nuclei that are not strongly magnetic, or very dilute samples.

The most modern NMR spectrometers use superconducting magnets, which can have field strengths as high as 14 Tesla and operate at 600 MHz. A superconducting

magnet is made of special alloys and must be cooled to liquid helium temperatures. The magnet is usually surrounded by a Dewar flask (an insulated chamber) containing liquid helium; in turn, this chamber is surrounded by another one containing liquid nitrogen. Instruments operating at frequencies above 100 MHz have superconducting magnets. NMR spectrometers with frequencies of 200 MHz, 300 MHz, and 400 MHz are now common in chemistry; instruments with frequencies of 900 MHz are used for special research projects.

3.8 CHEMICAL EQUIVALENCE—A BRIEF OVERVIEW

All the protons that give a single absorbance peak are called as Chemically Equivalent protons.

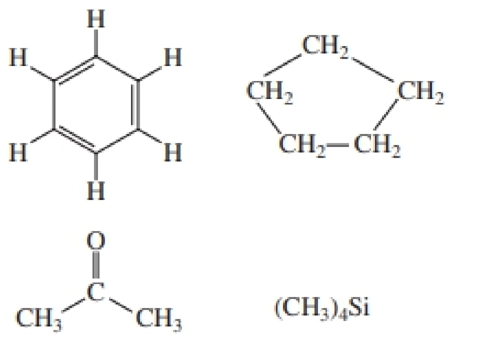
All of the protons found in **chemically identical environments** within a molecule are **chemically equivalent**, and they often exhibit the **same chemical shift.** Thus, all the protons in tetramethylsilane (TMS) or all the protons in benzene, cyclopentane, or acetone—which are molecules that have protons that are equivalent by symmetry considerations—have resonance at a single value. Each such compound gives rise to a single absorption peak in its NMR spectrum. The protons are said to be **chemically equivalent.**

2-**On the other hand, a molecule that has sets of protons that are chemically distinct from one another may give rise to a different absorption peak from each set, in which case the sets of protons are chemically nonequivalent.**

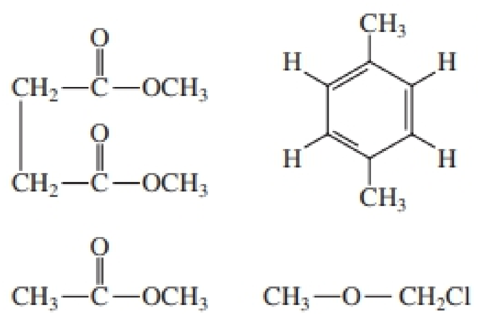
3- **It means that the number of peaks corresponds to the number of chemically distinct types of protons in the molecule.**

Molecules giving rise to two NMR absorption peaks—two different sets of chemically equivalent protons.

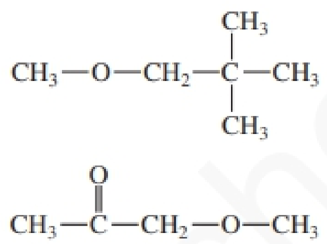
Molecules giving rise to three NMR absorption peaks—three different sets of chemically equivalent protons.



Molecules giving rise to one NMR absorption peak—all protons chemically equivalent



Molecules giving rise to two NMR absorption peaks—two different sets of chemically equivalent protons



Molecules giving rise to three NMR absorption peaks—three different sets of chemically equivalent protons

3.9 Integrals and Integration

The number of resonance peaks in NMR spectrum can tell us about the **number of different types of protons** in a molecule. NMR spectrum can also tell us that **how many protons of same type are present within the molecule**. In the NMR spectrum, the area under each peak is proportional to the number of hydrogens generating that peak.

Examples: Hence, in phenylacetone (see Fig. 3.12), the area ratio of the three peaks is 5:2:3, the same as the ratio of the numbers of the three types of hydrogens. The NMR spectrometer has the capability to electronically measure the area under each peak. It does this by tracing over each peak a vertically rising line, called the **integral**, which rises in height by an amount proportional to the area under the peak.

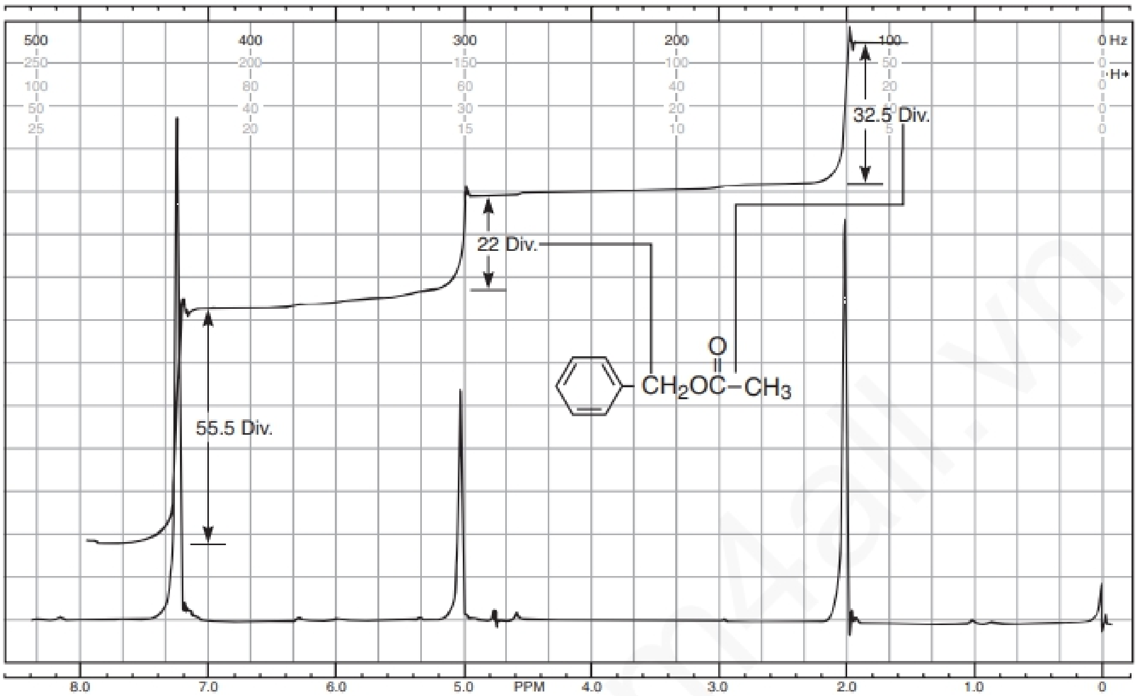


FIGURE 3.18 Determination of the integral ratios for benzyl acetate (60 MHz).

Figure 3.18 is a 60-MHz NMR spectrum of benzyl acetate, showing each of the peaks integrated in this way.

Note that the height of the integral line does not give the absolute number of hydrogens. It gives the relative number of each type of hydrogen. For a given integral to be of any use, there must be a second integral to which it may be referred. Benzyl acetate provides a good example of this. The first integral rises for 55.5 divisions on the chart paper; the second, 22.0 divisions; and the third, 32.5 divisions. These numbers are relative. One can find ratios of the types of protons by dividing each of the larger numbers by the smallest number:

$$\frac{55.5 \text{ div}}{22.0 \text{ div}} = 2.52 \qquad \frac{22.0 \text{ div}}{22.0 \text{ div}} = 1.00 \qquad \frac{32.5 \text{ div}}{22.0 \text{ div}} = 1.48$$

Thus, the number ratio of the protons of all the types is 2.52:1.00:1.48. If we arrive at the true ratio by multiplying each figure by 2 and rounding off to 5:2:3. Clearly, the peak at 7.3 ppm, which integrates for five protons, arises from the resonance of the aromatic ring protons, whereas that at 2.0 ppm, which integrates for three protons, is due to the methyl protons. The two-proton resonance at 5.1 ppm arises from the benzyl protons. Notice that the integrals give the simplest ratio, but not necessarily the true ratio, of numbers of protons of each type.

3.10 CHEMICAL ENVIRONMENT AND CHEMICAL SHIFT

If the resonance frequencies of all protons in a molecule were the same, NMR would be of little use to the organic chemist. Advantage of this technique is that **not only the different types of protons have different chemical shifts, but each also has a characteristic value of chemical shift**.

1-Every type of proton has only a limited range of delta values over which it gives resonance. Hence, the numerical value (in units or ppm) of the chemical shift for a proton gives a clue about the type of that proton, just as an infrared frequency gives a clue regarding the type of bond or functional group. For instance, notice that the **aromatic protons** of both phenylacetone (Fig. 3.12) and benzyl acetate (Fig. 3.18) **have resonance near 7.3 ppm**, and that both of the **methyl groups attached directly to a carbonyl have resonance at about 2.1 ppm**.

2-Aromatic protons characteristically have resonance near **7 to 8 ppm,** whereas **acetyl groups** (methyl groups of this type) have their resonance near **2 ppm.** These values of chemical shift are diagnostic. Notice also how the resonance of the benzyl (-CH_2-) protons comes at a higher value of chemical shift (5.1 ppm) in benzyl acetate than in phenylacetone (3.6 ppm). Being attached to the electronegative element oxygen, these protons are more deshielded (see Section 3.11) than those in phenylacetone. A trained chemist would readily recognize the probable presence of the oxygen from the value of chemical shift shown by these protons.

It is important to learn the ranges of chemical shifts over which the most common types of protons have resonance. Figure 3.20 is a correlation chart that contains the most essential and frequently encountered types of protons. Table 3.4 lists the chemical shift ranges for selected types of protons. For the beginner, it is often difficult to memorize a large body of numbers relating to chemical shifts and proton types. One actually need do this only crudely. It is more important to “get a feel” for the regions and the types of protons than to know a string of actual numbers. To do this, study Figure 3.20

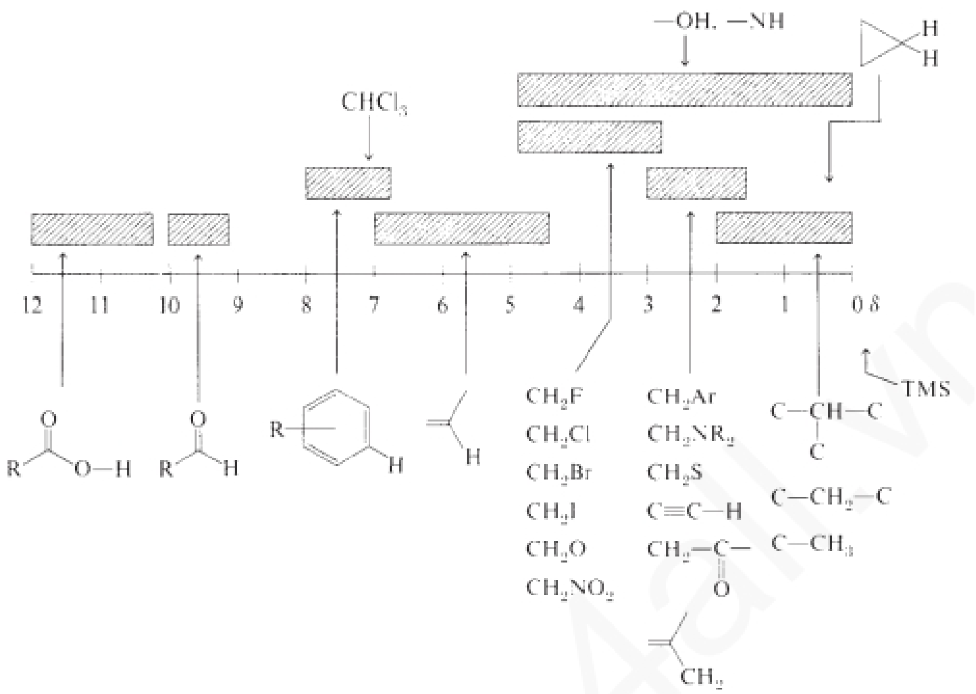


FIGURE 3.20 A simplified correlation chart for proton chemical shift values.

3.11 LOCAL DIAMAGNETIC SHIELDING

3.11.1- Electronegativity Effects:

The electronegativity trend is easy to explain.

1-The electron present around each proton are spinning and creating a local magnetic field. This local magnetic field protects all protons from the applied magnetic field in which our sample is present. This process is called **local diamagnetic shielding.**

2-If the electronegative element is attached to the carbon atom on which the proton is attached, electronegative element withdraws the valence electron density around that proton and the proton is de-shielded. In this way the proton feels more outer magnetic field and the chemical shift of that proton simply increases as the electronegativity of the attached element increases. Table 3.5 illustrates this relationship for several compounds of the type CH3X.

2-As the number of electrons withdrawing substituents increases, the deshielding effect becomes stronger.

3-With an increase in distance, the effect is also decreased. An electronegative element having little effect on protons that are more than three carbons distant. **Table 3.6** illustrates these effects for the underlined protons.

TABLE 3.5
DEPENDENCE OF THE CHEMICAL SHIFT OF CH_3X ON THE ELEMENT X

Compound CH_3X	CH_3F	CH_3OH	CH_3Cl	CH_3Br	CH_3I	CH_4	$(CH_3)_4Si$
Element X	F	O	Cl	Br	I	H	Si
Electronegativity of X	4.0	3.5	3.1	2.8	2.5	2.1	1.8
Chemical shift δ	4.26	3.40	3.05	2.68	2.16	0.23	0

TABLE 3.6
SUBSTITUTION EFFECTS

$C\underline{H}Cl_3$	$C\underline{H}_2Cl_2$	$C\underline{H}_3Cl$	$-C\underline{H}_2Br$	$-C\underline{H}_2-CH_2Br$	$-C\underline{H}_2-CH_2CH_2Br$
7.27	5.30	3.05	3.30	1.69	1.25

4-Substituents that have this type of effect are said to deshield the proton. The greater the electronegativity of the substituent, the more it Deshields protons and hence the greater is the chemical shift of those protons.

It is important to learn the ranges of chemical shifts over which the most common types of protons have resonance. Figure 3.20 is a correlation chart that contains the most essential and frequently encountered types of protons.

Table 3.4 lists the chemical shift ranges for selected types of protons. For the beginner, it is often difficult to memorize a large body of numbers relating to chemical shifts and proton types. One actually need do this only crudely. It is more important to "get a feel" for the regions and the types of protons than to know a string of actual numbers. To do this,study Figure 3.20 carefully. Table 3.4 and Appendix 2 give more detailed listings of chemical shifts.

3.11.2 Hybridization Effect

The hybridization of the carbon, to which proton is attached, play an important role in the chemical shift of the protons.

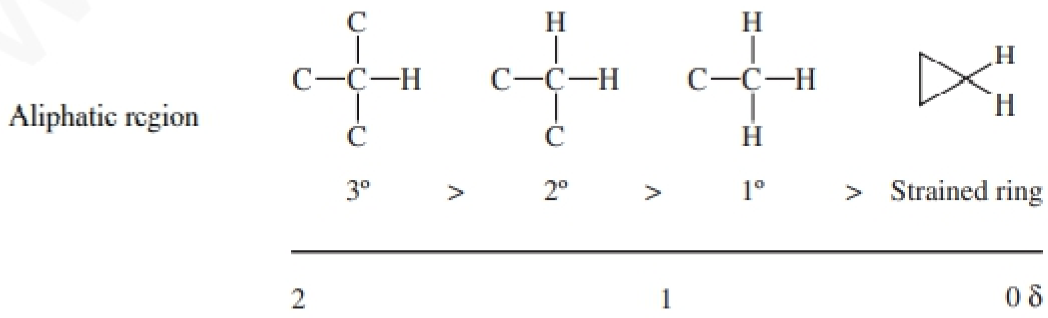
A- **sp³ Hydrogens** : Notice that all hydrogens attached to purely sp^3 carbon atoms (ClCH$_3$, C-CH$_2$-C, C-CH-C$_2$, cycloalkanes) have resonance in the limited range from 0 to 2 ppm, provided that no electronegative elements or -bonded groups are nearby. At the extreme right of this range are TMS (0 ppm) and

TABLE 3.4
APPROXIMATE CHEMICAL SHIFT RANGES (PPM) FOR SELECTED TYPES OF PROTONS[a]

Type of proton		δ (ppm)
$R-CH_3$		0.7 – 1.3
$R-CH_2-R$		1.2 – 1.4
R_3CH		1.4 – 1.7
$R-C=C-C-H$		1.6 – 2.6
$R-C(=O)-C-H$, $H-C(=O)-C-H$		2.1 – 2.4
$RO-C(=O)-C-H$, $HO-C(=O)-C-H$		2.1 – 2.5
$N\equiv C-C-H$		2.1 – 3.0
C_6H_5-C-H		2.3 – 2.7
$R-C\equiv C-H$		1.7 – 2.7
$R-S-H$	var	1.0 – 4.0[b]
$R-N-H$	var	0.5 – 4.0[b]
$R-O-H$	var	0.5 – 5.0[b]
C_6H_5-O-H	var	4.0 – 7.0[b]
C_6H_5-N-H	var	3.0 – 5.0[b]
$R-C(=O)-N-H$	var	5.0 – 9.0[b]
$R-N-C-H$		2.2 – 2.9
$R-S-C-H$		2.0 – 3.0
$I-C-H$		2.0 – 4.0
$Br-C-H$		2.7 – 4.1
$Cl-C-H$		3.1 – 4.1
$R-S(=O)_2-O-C-H$		ca. 3.0
$RO-C-H$, $HO-C-H$		3.2 – 3.8
$R-C(=O)-O-C-H$		3.5 – 4.8
O_2N-C-H		4.1 – 4.3
$F-C-H$		4.2 – 4.8
$R-C=C-H$		4.5 – 6.5
C_6H_5-H		6.5 – 8.0
$R-C(=O)-H$		9.0 – 10.0
$R-C(=O)-OH$		11.0 – 12.0

hydrogens attached to carbons in highly strained rings (0–1 ppm)—as occurs, for example, with cyclopropyl hydrogens. Most methyl groups occur near 1 ppm if they are attached to other sp^3 carbons. **Methylene-group** hydrogens (attached to sp^3 carbons) appear at greater chemical shifts (near

1.**2 to 1.4** ppm) than do methyl-group hydrogens. **Tertiary methine hydrogens** occur at higher chemical shift than secondary hydrogens, which in turn have a greater chemical shift than do primary or methyl hydrogens. The following diagram illustrates these relationships:



B-**Sp² Hydrogens**

1-Simple vinyl hydrogens (-CJ=C-H) have resonance in the range from **4.5 to 7** ppm. In an sp^2-s C-H bond, the carbon atom has more **s** character (33% s). As s orbital held electron closer to nucleus, making it "more electronegative" than an sp**3** carbon (25% s). Thus, **vinyl hydrogens have a greater chemical shift (5 to 6 ppm) than aliphatic hydrogens on sp3 carbons (1 to 4 ppm).**

2-**Aromatic hydrogens appear in a range farther downfield (7 to 8 ppm).** The downfield positions of vinyl and aromatic resonances are, however, greater than one would expect based on these hybridization differences. Another effect, called **anisotropy**, is responsible for the largest part of these shifts (and will be discussed next).

3-**Aldehyde protons (also attached to sp^2 carbon) appear even farther downfield (9 to 10 ppm) than aromatic protons** since the inductive effect of the electronegative oxygen atom further decreases the electron density on the attached proton. Aldehyde protons, like aromatic and alkene protons, exhibit an anomalously large chemical shift due to **anisotropy.**

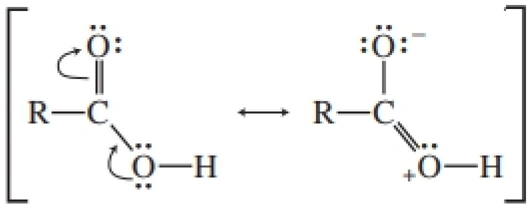
C-**SP Hydrogens:**

Acetylenic hydrogens appear at 2 to 3 ppm, lower than sp^2 hybridized .
It is due to anisotropy (to be discussed ahead). On the basis of hybridization alone, as already discussed, one would expect the acetylenic proton to have a chemical shift greater than that of the vinyl proton. An **sp** carbon should

behave as if it were more electronegative than an sp^2 carbon. This is the opposite of what is actually observed.

3.11.3 Acidic and exchangeable protons; Hydrogen bonding

Some of the least de-shielded protons. These appear at 10-12 ppm.



Both resonance and the electronegativity effect of oxygen withdraw electrons from the acid proton.

Hydrogen Bonding and Exchangeable Hydrogens: Protons that can exhibit hydrogen bonding (e.g., hydroxyl or amino protons) exhibit extremely variable absorption positions over a wide range. They are usually found attached to a heteroatom. Table 3.7 lists the ranges over which some of these types of protons are found.

TABLE 3.7
TYPICAL RANGES FOR PROTONS WITH VARIABLE CHEMICAL SHIFT

Acids	RCOOH	10.5–12.0 ppm
Phenols	ArOH	4.0–7.0
Alcohols	ROH	0.5–5.0
Amines	RNH_2	0.5–5.0
Amides	$RCONH_2$	5.0–8.0
Enols	CH=CH–OH	>15

The more hydrogen bonding that takes place, the more deshielded a proton becomes. The amount of hydrogen bonding is often a function of concentration and temperature. **The more concentrated the solution, the more molecules can come into contact with each other and hydrogen bond.** At high dilution (no H bonding), hydroxyl protons absorb near **0.5–1.0** ppm; in concentrated solution, their absorption is closer to **4–5 ppm**. Protons on other heteroatoms show similar tendencies. Hydrogens that can exchange either with the solvent medium or with one another also tend to be variable in their absorption positions.

3.12 MAGNETIC ANISOTROPY

The chemical resonance of protons of vinyl, acetylene, aldehyde, benzene and other aromatic systems are not easy to explain by only using electron-withdrawing or hybridization effects.

In each of these cases, the **anomalous shift is due to the presence of an unsaturated system in the vicinity of the proton** in question.

1-Benzene: Take benzene, for example. When it is placed in a magnetic field, the p electrons in the aromatic ring system are induced to circulate around the ring. This circulation is called a **ring current.**

The moving electrons generate a **magnetic fie**ld much like that generated in a loop of wire through which a current is induced to flow.

This magnetic field produced by the circulation of p electrons is not uniform throughout the ring as can be seen in Fig 3,21. This phenomenon is called **diamagnetic anisotropy**.

The direction of this diamagnetic anisotropy, in the periphery, is same as the direction of outer applied magnet. So, the protons of the benzene ring will feel more applied field and are said to be de-shielded by the diamagnetic anisotropy of the ring. So, appear at greater chemical shift than expected.

However, in the center of the ring the magnetic anisotropic effect is opposite to that of applied magnet and will shield the center protons(if any).

If a proton were placed in the center of the ring rather than on its periphery, it would be found to be shielded since the field lines there would have the opposite direction from those at the periphery

B-**Acetylene**: In acetylene, the magnetic field generated by induced circulation of the p electrons has a geometry such that the acetylenic hydrogens are shielded (Fig. 3.22). Hence, acetylenic hydrogens have resonance at lower chemical shift than expected. Figure 3.24 shows the effects of anisotropy in several actual molecules.

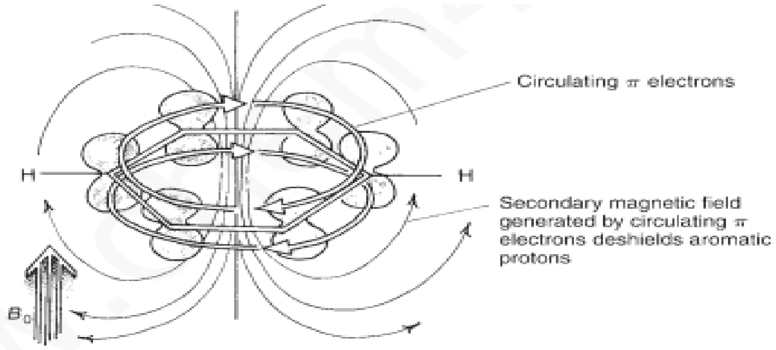


FIGURE 3.21 Diamagnetic anisotropy in benzene.

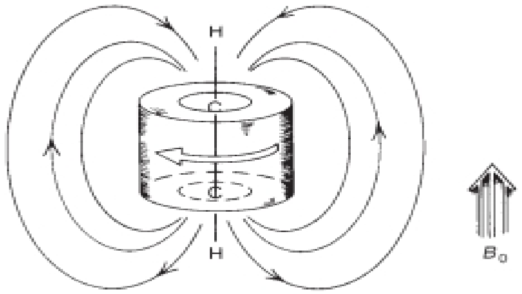


FIGURE 3.22 Diamagnetic anisotropy in acetylene.

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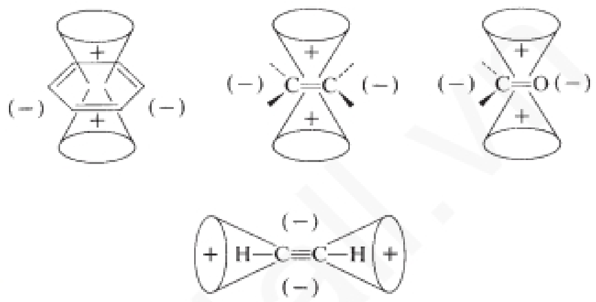
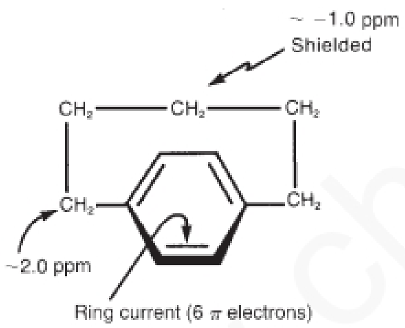
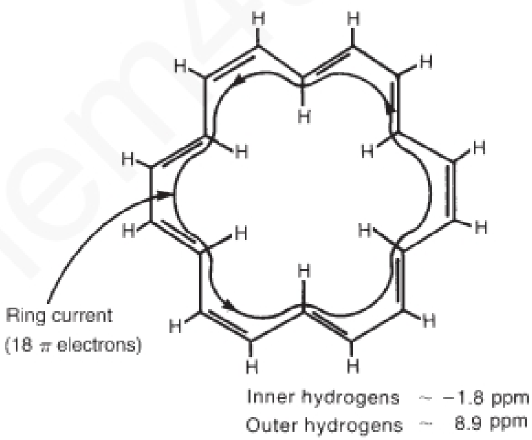


FIGURE 3.23 Anisotropy caused by the presence of π electrons in some common multiple-bond systems.





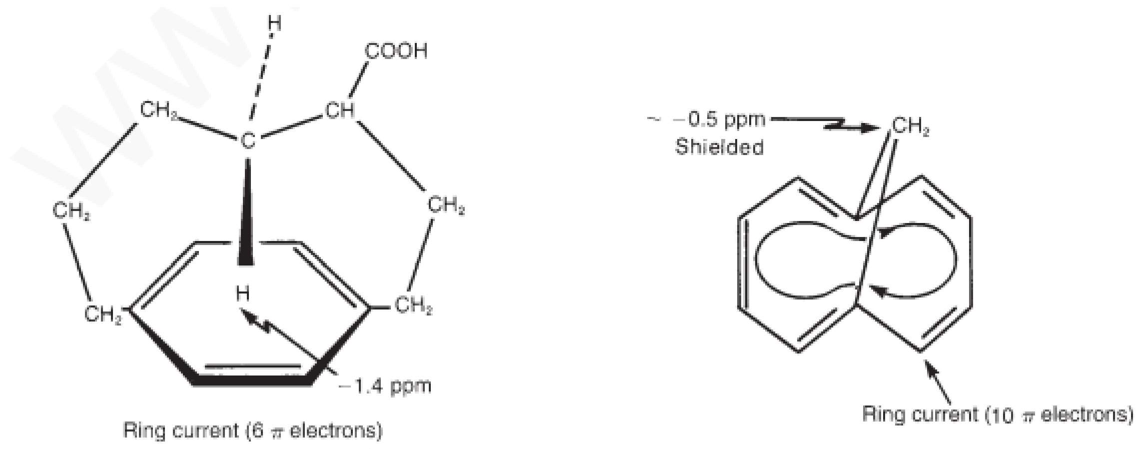
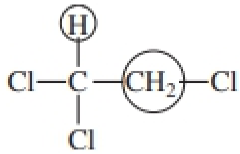


FIGURE 3.24 The effects of anisotropy in some actual molecules.

3.13 SPIN–SPIN SPLITTING (n +1) RULE

In most of the cases, even in simple molecules, proton does not give a single resonance peak. For instance, in **1,1,2-trichloroethane**, there are two chemically distinct types of hydrogens:



On the basis of the information given thus far, one would predict two resonance peaks in the NMR spectrum of 1,1,2-trichloroethane, with an area ratio (integral ratio) of 2:1.

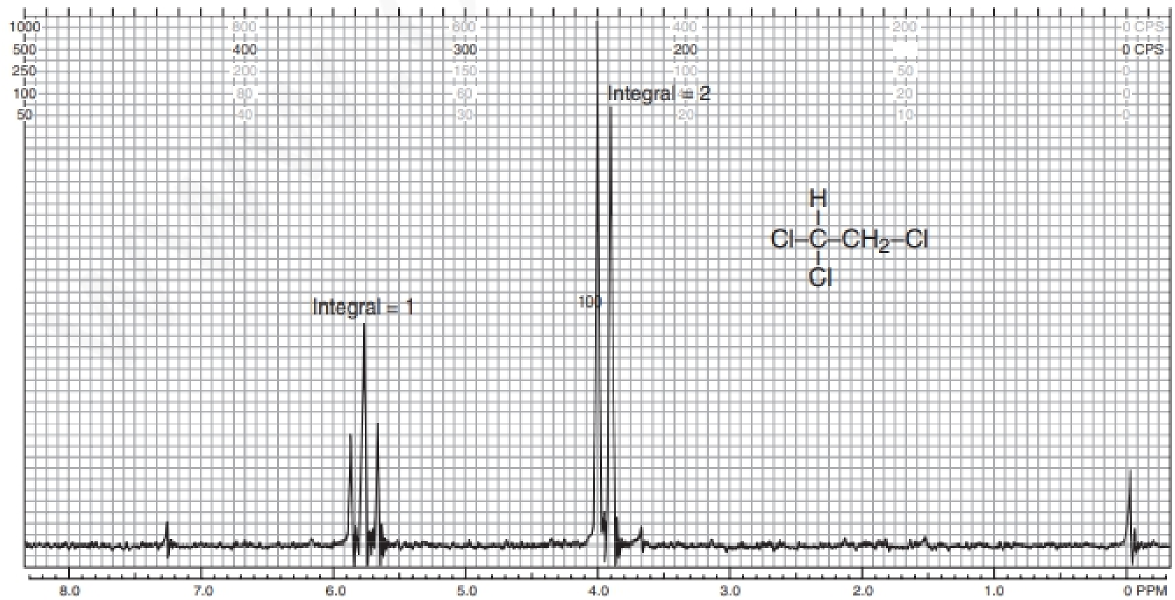


FIGURE 3.25 The ^{1}H NMR spectrum of 1,1,2-trichloroethane (60 MHz).

In reality, the high-resolution NMR spectrum of this compound has five peaks: a group of three peaks (called a triplet) at 5.77 ppm and a group of two peaks (called a doublet) at 3.95 ppm. **Figure 3.25 shows this spectrum**.

The methine (CH) resonance (5.77 ppm) is said to be **split into a triplet**, and **the methylene resonance** (3.95 ppm) is **split into a doublet**. The area under the three triplet peaks is 1, relative to an area of 2 under the two doublet peaks. **This phenomenon, called spin–spin splitting,** can be explained empirically by the so-called **n + 1** Rule.

Each type of proton "senses" the number of equivalent protons (n) on the adjacent carbon atom(s), and its resonance peak is split into (n + 1) components. Here **n** is the number of protons on adjacent carbon(s). Examine the case at hand, 1,1,2-trichloroethane, utilizing the n + 1 Rule.

First the **lone methine hydrogen** is situated next to a carbon bearing two methylene protons. According to the rule, it has two **equivalent neighbors (n = 2) and is split into n + 1 = 3 peaks (a triplet)**.

The methylene protons are situated next to a carbon bearing only one methine hydrogen. According to the rule, these protons have one neighbor (n = 1) and are split into n + 1 = 2 peaks (a doublet).

Equivalent protons behave as a group

Two neighbors give a triplet
($n + 1 = 3$) (area = 1)

One neighbor gives a doublet
($n + 1 = 2$) (area = 2)

Example 2: Figure 3.26 is the spectrum of ethyl iodide (CH_3CH_2I). Notice that the methylene protons are split into a quartet (four peaks), and the methyl group is split into a triplet (three peaks). This is explained as follows:

Three equivalent neighbors give a quartet
($n + 1 = 4$) (area = 2)

Two equivalent neighbors give a triplet
($n + 1 = 3$) (area = 3)

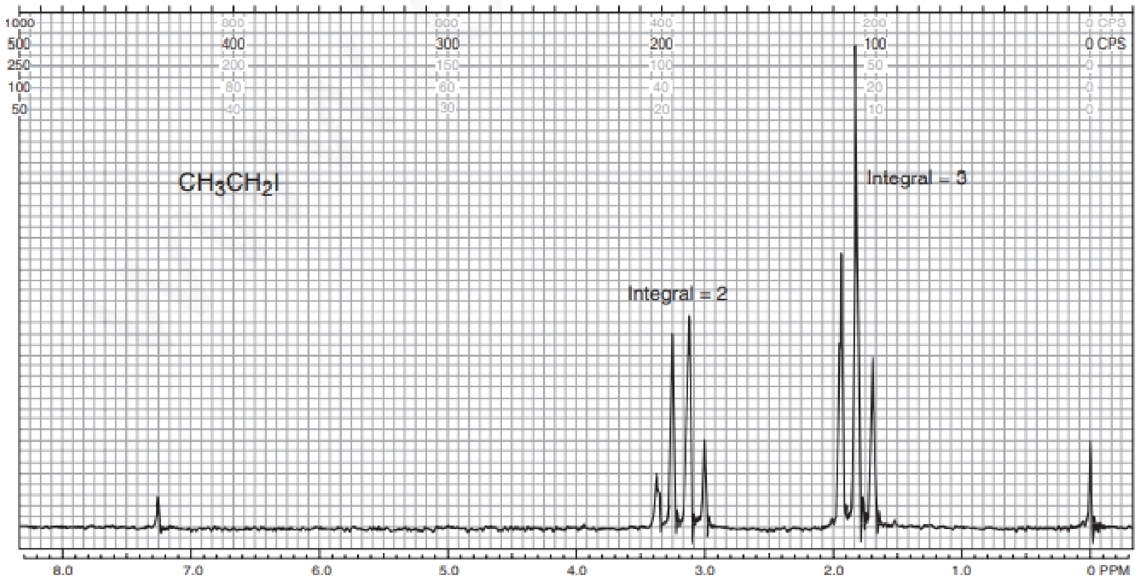
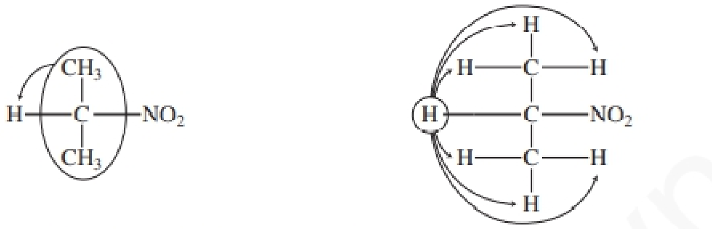


FIGURE 3.26 The ^{1}H NMR spectrum of ethyl iodide (60 MHz).

Example 3: Finally, consider 2-nitropropane, which has the spectrum given in Figure 3.27.



One neighbor gives a doublet
($n + 1 = 2$) (area = 6)

Six equivalent neighbors give a septet
($n + 1 = 7$) (area = 1)

Notice that in the case of 2-nitropropane there are two adjacent carbons that bear hydrogens (two carbons, each with three hydrogens), and that all six hydrogens as a group split the methine hydrogen into a septet.

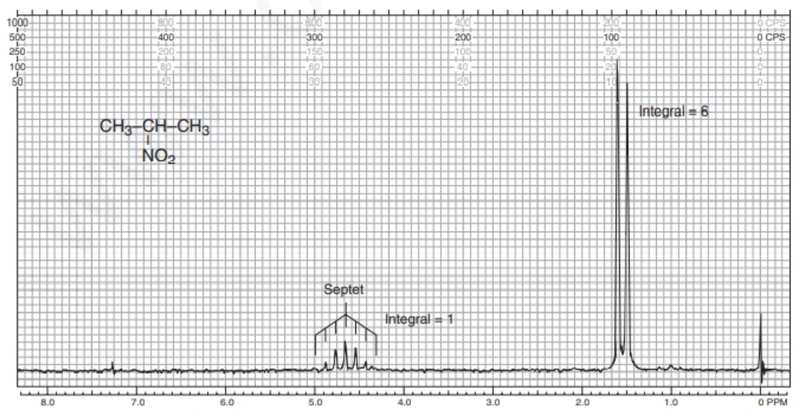


FIGURE 3.27 The ^{1}H NMR spectrum of 2-nitropropane (60 MHz).

From the rules of electronegativity and hybridization, one can predict that which carbon show resonance at which chemical shift. Thus, in 1,1,2-trichloroethane, the methine hydrogen (on a carbon bearing two Cl atoms) has a larger chemical shift than the methylene protons (on a carbon bearing only one Cl atom).

In ethyl iodide, the hydrogens on the carbon-bearing iodine have a larger chemical shift than those of the methyl group.

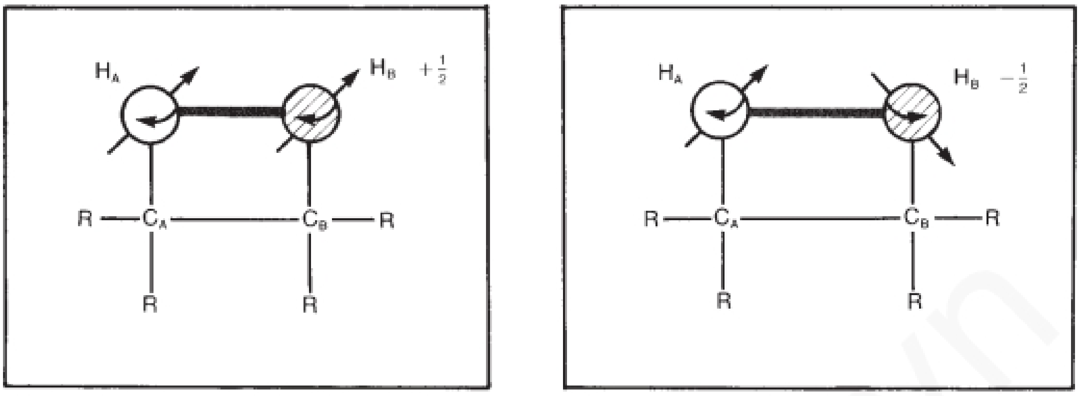
In 2-nitropropane, the methine proton (on the carbon bearing the nitro group) has a larger chemical shift than the hydrogens of the two methyl groups.

Finally, the spin–spin splitting gives information that how many hydrogens are adjacent to each type of hydrogen that is giving an absorption peak or, as in these cases, an absorption multiplet.

For reference, some commonly encountered spin–spin splitting patterns are collected in Table 3.8.

3.14 THE ORIGIN OF SPIN–SPIN SPLITTING

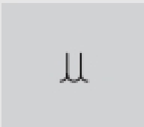
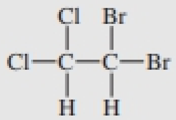
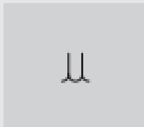
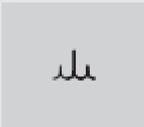
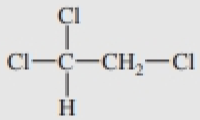
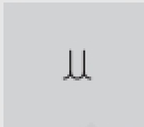
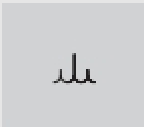
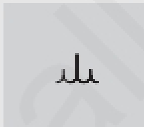
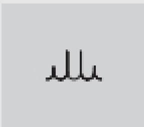
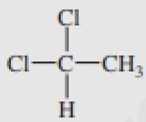
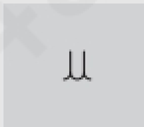
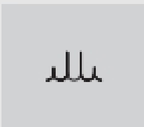
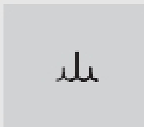
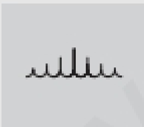
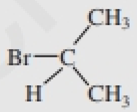
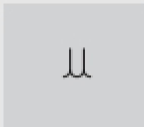
Spin–spin splitting arises because hydrogens on adjacent carbon atoms can "**sense" one another**. The hydrogen on carbon A can sense the spin direction of the hydrogen on carbon B. In some molecules of the solution, the hydrogen on carbon B has spin + 1/2 (X-type molecules); in other molecules of the solution, the hydrogen on carbon B has spin – 1/2 (Y-type molecules).



X-type molecule Y-type molecule

FIGURE 3.28 Two different molecules in a solution with differing spin relationships between protons H_A and H_B.

TABLE 3.8
SOME EXAMPLES OF COMMONLY OBSERVED SPLITTING PATTERNS IN COMPOUNDS

1H				1H
1H				2H
2H		$Cl—CH_2—CH_2—Br$		2H
1H				3H
2H		$Cl—CH_2—CH_3$		3H
1H				6H
	Downfield		Upfield	

The **chemical shift of proton A is influenced** by the **direction of the spin in proton B**. Proton A is said to be coupled to proton B. Its magnetic environment is affected by whether proton B has a +1/2 or a – 1/2 spin state. Thus, proton **A** absorbs at a slightly different chemical shift value in type **X** molecules than in type **Y** molecules. In fact, **in X-type molecules, proton A is slightly deshielded** because the field of **proton B is aligned with the applied field,** and its magnetic moment adds to the applied field. In **Y-type molecules, proton A is slightly shielded** with respect to what its chemical shift would be in the absence of coupling. In this latter case, the field of proton **B reduces the effect of the applied field on proton A**.

Since in a given solution there are approximately equal numbers of X- and Y-type molecules at any given time, two absorptions of nearly equal intensity are observed for proton A. The resonance of proton A is said to have been split by proton B, and the general phenomenon is called **spin–spin splitting**. Figure 3.29 summarizes the spin–spin splitting situation for proton A.

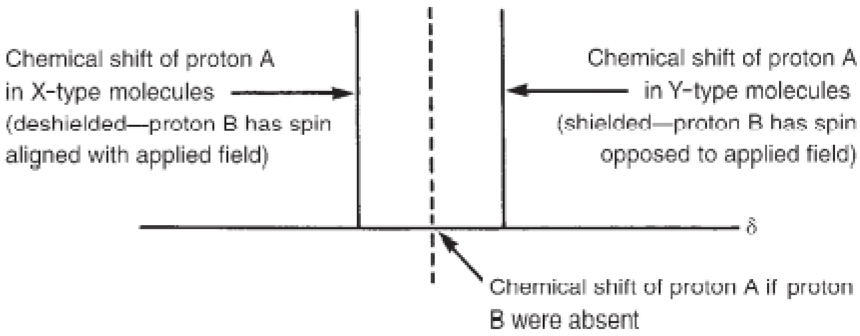
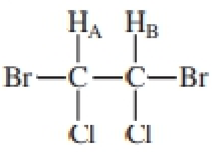
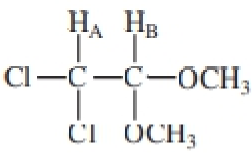


FIGURE 3.29 The origin of spin–spin splitting in proton A's NMR spectrum.

Of course, proton A also "splits" proton B since proton A can adopt two spin states as well. The final spectrum for this situation consists of two doublets:

Two doublets will be observed in any situation of this type except one in which protons A and B are identical by symmetry, as in the case of the first of the following molecules:





The first molecule would give only a single NMR peak since protons A and B have the same chemical shift value and are, in fact, identical.

The second molecule would probably exhibit the two-doublet spectrum since protons A and B are not identical and would surely have different chemical shifts.

Note that except in unusual cases, coupling (spin–spin splitting) occurs only between hydrogens on adjacent carbons. Hydrogens on nonadjacent carbon atoms generally do not couple strongly enough to produce observable splitting, although there are some important exceptions to this generalization, which will be further discussed.

3.15 THE ETHYL GROUP (CH_3CH_2-)

Now consider ethyl iodide, which has the spectrum shown in Figures 3.26 and 3.30. **The methyl protons give rise to a triplet** centered at 1.83 ppm, and the **methylene protons give a quartet** centered at 3.20 ppm. This pattern and the relative intensities of the component peaks can be explained with the use of the model for the two-proton case outlined in Section 3.13.

First, look at the methylene protons and their pattern, which is a quartet. The methylene protons are split by the methyl protons, and to understand the splitting pattern, you must examine the **various possible spin arrangements of the protons for the methyl group**, which are shown in Figure 3.31. Some of the **eight possible spin arrangements are identical to each other** since one methyl proton is indistinguishable from another and since there is free rotation in a methyl group.

Out of these eight arrangements, one possibility is when all three protons are spinning in clock wise direction. Second possibility is when all three protons are spinning in anti-clock wise. However, there are six possibilities that all three protons are spinning in different direction, three for clock wise and three for anti-clock wise.

Due to this reason, intensity ratios are 1:3:3:1. Each of these different spin arrangements of the methyl protons (except the sets of degenerate ones, which are effectively identical) gives the methylene protons in that molecule a different chemical shift value.

Now consider the case of triplet for methyl protons. Figure 3.32 provides a similar analysis of the methyl splitting pattern, showing the four possible spin arrangements of the methylene protons, one in which all two protons are spinning clock wise, second in which all two protons are spinning anti-clock wise and two cases when both two protons are spinning differently from each other. Examination of this figure makes it easy to explain the origin of the triplet for the methyl group and the intensity ratios of 1:2:1. Now one can see the origin of the ethyl pattern and the explanation of its intensity ratios.

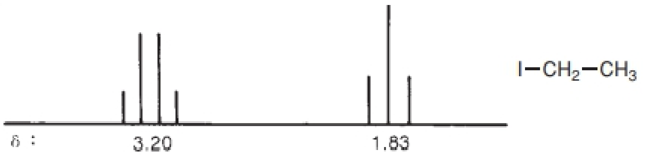


FIGURE 3.30 The ethyl splitting pattern.

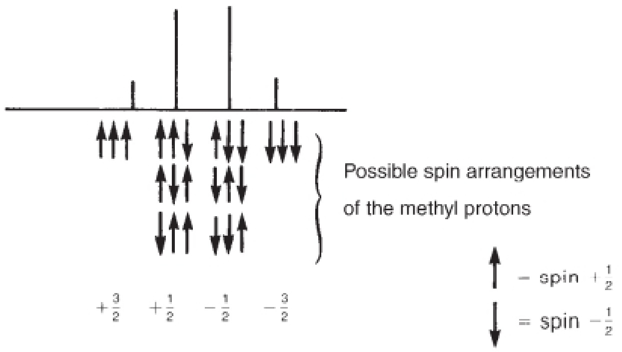


FIGURE 3.31 The splitting pattern of methylene protons due to the presence of an adjacent methyl group.

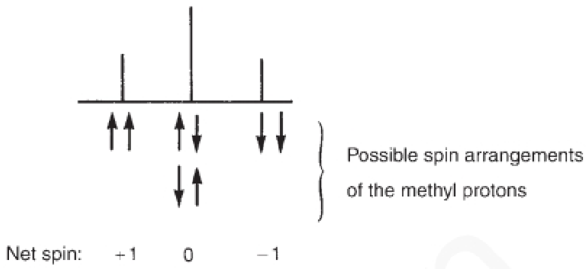


FIGURE 3.32 The splitting pattern of methyl protons due to the presence of an adjacent methylene group.

The occurrence of spin–spin splitting is very important for the organic chemist as it gives additional structural information about molecules. In this way we can understand the number of protons on neighboring carbons. 1-From the chemical shift one can determine what type of proton is being split, and

2-from the integral (the area under the peaks) one can determine the relative numbers of the types of hydrogen.

This is a great amount of structural information, and it is invaluable to the chemist attempting to identify a particular compound.

3.16 PASCAL'S TRIANGLE

With the help of Pascal's triangle, one can predict the intensity ratios of multiplets derived from the n + 1 Rule (Fig. 3.33).

Each entry in the triangle is the sum of the two entries above it and to its immediate left and right.

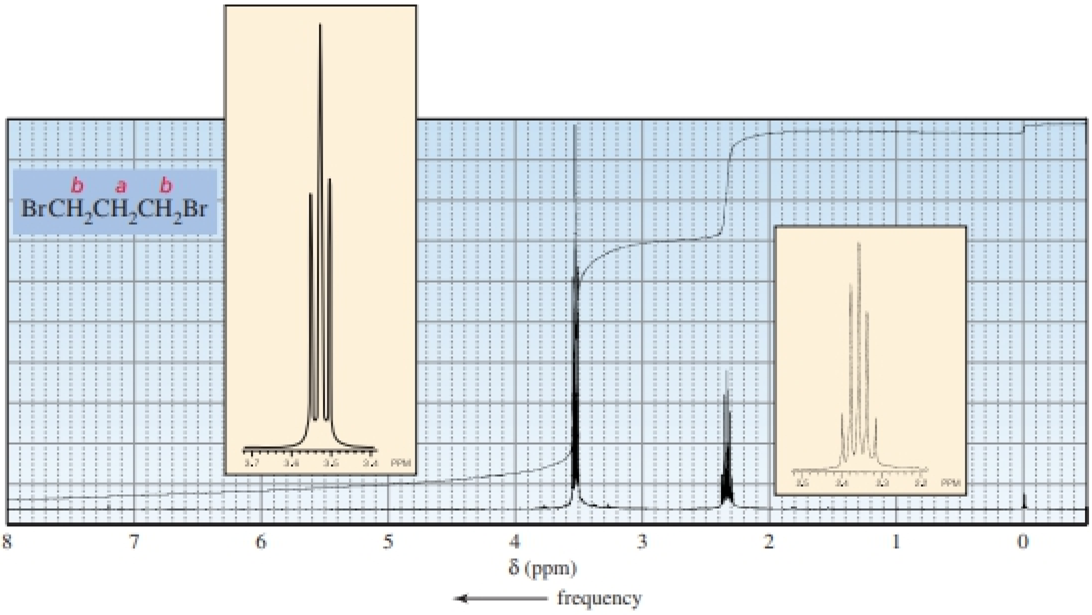
Singlet	1
Doublet	1 1
Triplet	1 2 1
Quartet	1 3 3 1
Quintet	1 4 6 4 1
Sextet	1 5 10 10 5 1
Septet	1 6 15 20 15 6 1

FIGURE 3.33 Pascal's triangle.

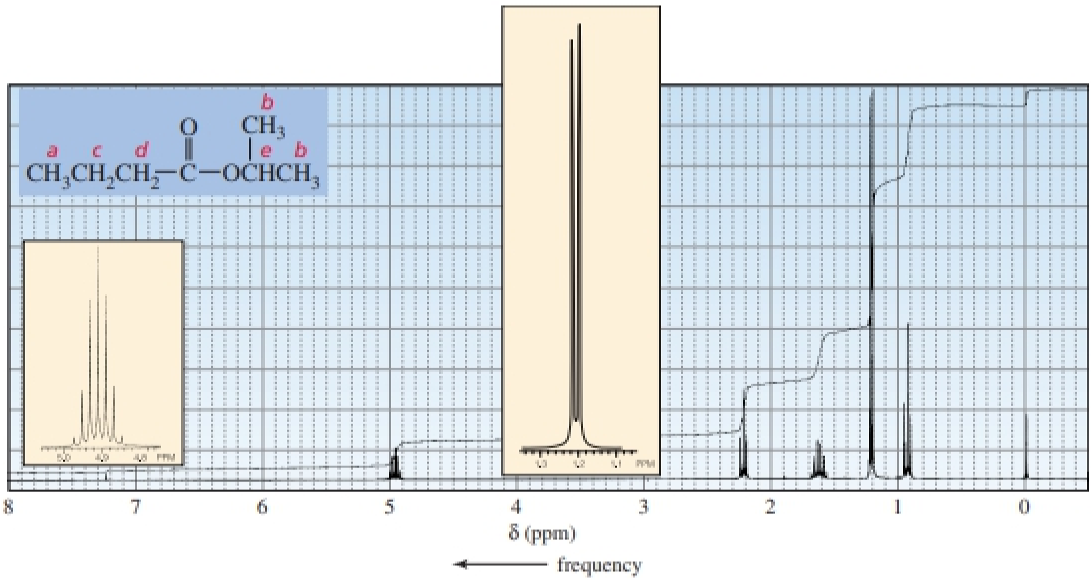
3.17 More Examples of NMR Spectra

The ^{1}H NMR spectrum of bromomethane shows one singlet. The three methyl protons are chemically equivalent, and chemically equivalent protons do not split each other's signal. The four protons in 1,2-dichloroethane are also chemically equivalent, so its ^{1}H NMR spectrum also shows one singlet.

There are two signals in the NMR spectrum of 1,3-dibromopropane (Figure 14.15). The signal for the protons is split into a triplet by the two hydrogens on the adjacent carbon. The protons have two adjacent carbons that are bonded to protons. The protons on one adjacent carbon are equivalent to the protons on the other adjacent carbon. Because the two sets of protons are equivalent, the rule is applied to both sets at the same time. In other words, N is equal to the sum of the equivalent protons on both carbons. So the signal for the protons is split into a quintet Integration confirms that two methylene groups contribute to the signal because twice as many protons give rise to the signal as to the signal. Hb Ha Hb



The NMR spectrum of isopropyl butanoate shows five signals (Figure 14.16). The signal for the protons is split into a triplet by the protons. The signal for the protons is split into a doublet by the proton. The signal for the protons is split into a triplet by the protons, and the signal for the proton is split into a septet by the protons. The signal for the protons is split by both the and protons. Because the and protons are not equivalent, the rule has to be applied separately to each set. Thus, the signal for the protons will be split into a



quartet by the protons, and each of these four peaks will be split into a triplet by the protons: As a result, the signal for the protons is a multiplet (a signal that is more complex that a triplet, quartet, quintet, etc.). The reason we do not see 12 peaks is that some of them overlap (Section 14.13).

The NMR spectrum of 3-bromo-1-propene shows four signals (Figure 14.17). Although the and protons are bonded to the same carbon, they are not chemically equivalent (one is cis to the bromomethyl group, the other is trans to the bromomethyl group), so each produces a separate signal. The signal for the protons is split into a doublet by the proton. Notice that the signals for the three vinylic protons are at relatively high frequencies because of diamagnetic anisotropy (Section 14.9). The signal for the proton is a multiplet because it is split separately by the and protons.

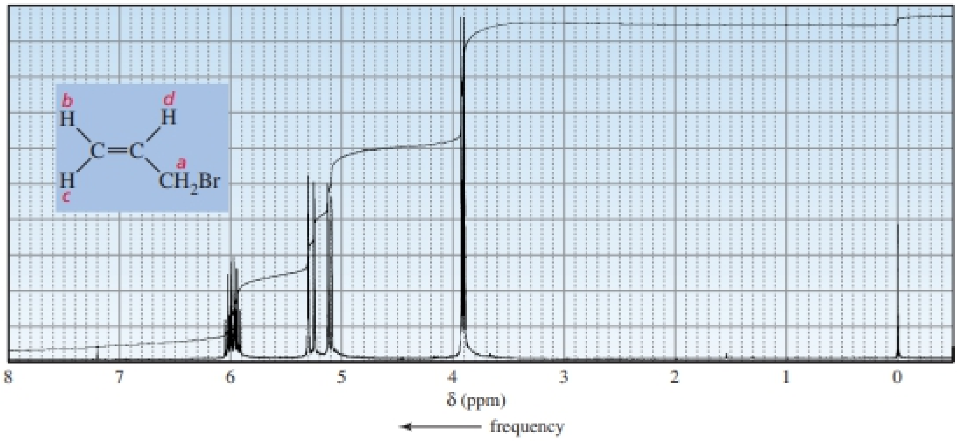
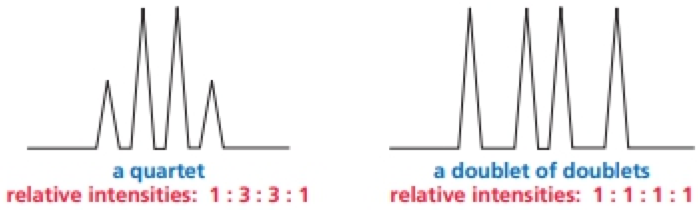


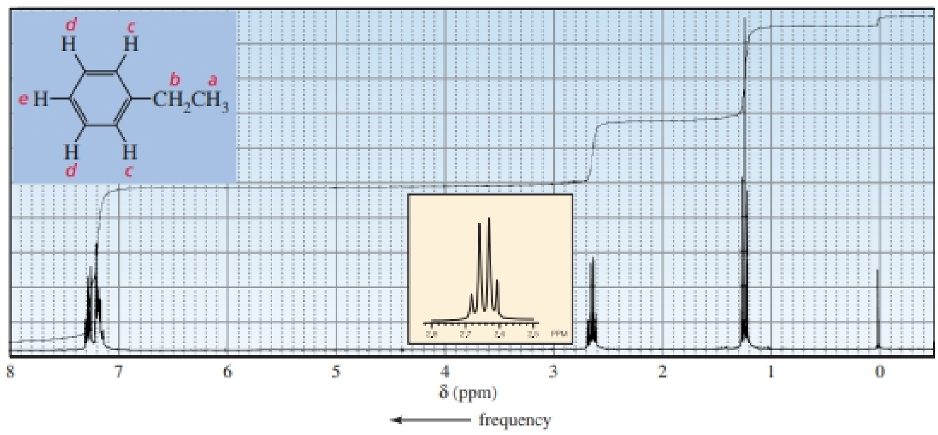
Figure 14.17 ^{1}H NMR spectrum of 3-bromo-1-propene.

Because the and protons are not equivalent, they split one another's signal. This means that the signal for the proton is split into a doublet by the proton and that each of the peaks in the doublet is split into a doublet by the proton. The signal for the proton should be what is called a doublet of doublets. The signal for the proton should also be a doublet of doublets. However, the mutual splitting of the signals of two nonidentical protons bonded to the same carbon, caused by what is called geminal coupling, is often too small to be observed (Section 14.12). Therefore, the signals for the and protons in Figure 14.17 appear as doublets rather than as doublets of doublets. (If the signals were expanded, the doublet of doublets would be observed.) Notice the difference between a quartet and a doublet of doublets. Both have four peaks. A quartet results from splitting by three equivalent adjacent protons; it has relative peak intensities of and the individual peaks are equally spaced. A doublet of doublets, on the other hand, results from splitting by two nonequivalent adjacent protons; it has relative peak intensities of and the individual peaks are not necessarily equally spaced (see Figure 14.24).



There are five sets of chemically equivalent protons in ethylbenzene (Figure 14.18). We see the expected triplet for the protons and the quartet for the protons. (This is a characteristic pattern for an ethyl group.) We expect the signal for the protons to be a doublet and the signal for the proton to be a triplet. Because the and protons are not equivalent, they must be considered separately in determining the splitting of the signal for the protons. Therefore, we expect the signal for the protons to be split into a doublet by the protons and each peak of the doublet to be split into another doublet by the proton, forming a doublet of doublets. However, we do not see three distinct signals for the and protons in Figure 14.18. Instead, we see overlapping signals. Apparently, the electronic effect (i.e., the electron-donating/electronwithdrawing ability) of an ethyl substituent is not sufficiently different from that of a hydrogen to cause a difference in the environments of the and protons that is large enough to allow them to appear as separate signals.

Figure 14.18 ▶ ^1H NMR spectrum of ethylbenzene. The signals for the H_c, H_d, and H_e protons overlap.



In contrast to the and protons of ethylbenzene, the and protons of nitrobenzene show three distinct signals (Figure 14.19), and the multiplicity of each signal is what we predicted for the signals for the benzene ring protons in

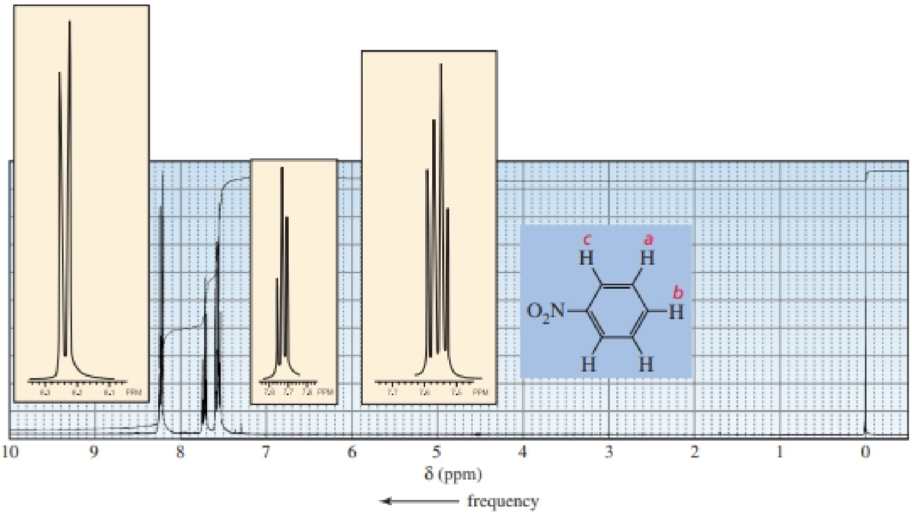


Figure 14.19 ▶ ^{1}H NMR spectrum of nitrobenzene. The signals for the H_a, H_b, and H_c protons do not overlap.

ethylbenzene (is a doublet, is a triplet, and is a doublet of doublets). The nitro group is sufficiently electron withdrawing to cause the and protons to be in different enough environments that their signals do not overlap. Notice that the signals for the benzene ring protons in Figures 14.18 and 14.19 occur in the 7.0–8.5 ppm region. Other kinds of protons usually do not resonate in this region, so signals in this region of an spectrum indicate that the compound probably contains an aromatic ring.

3.17 THE COUPLING CONSTANT

The distance between the peaks in a simple multiplet is called the coupling constant J.

The coupling constant is a measure of how strongly a nucleus is affected by the spin states of its neighbor. The spacing between the multiplet peaks is measured on the same scale as the chemical shift, and the coupling constant is always expressed in Hertz (Hz). In ethyl iodide, for instance, the coupling constant J is 7.5 Hz.

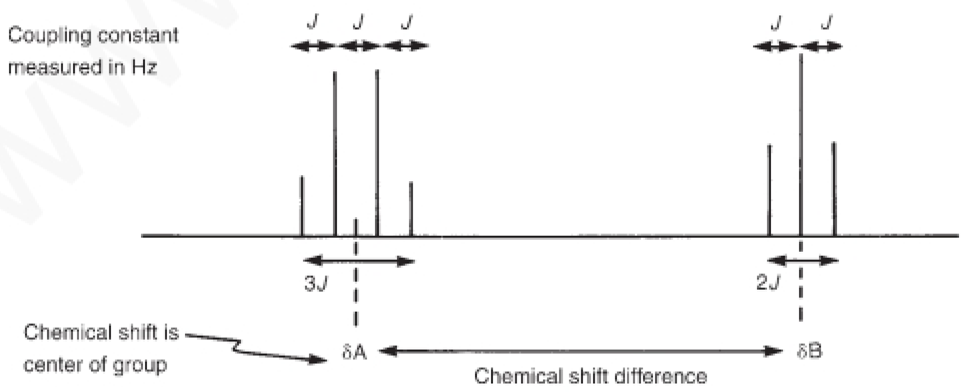


FIGURE 3.34 The definition of the coupling constants in the ethyl splitting pattern.

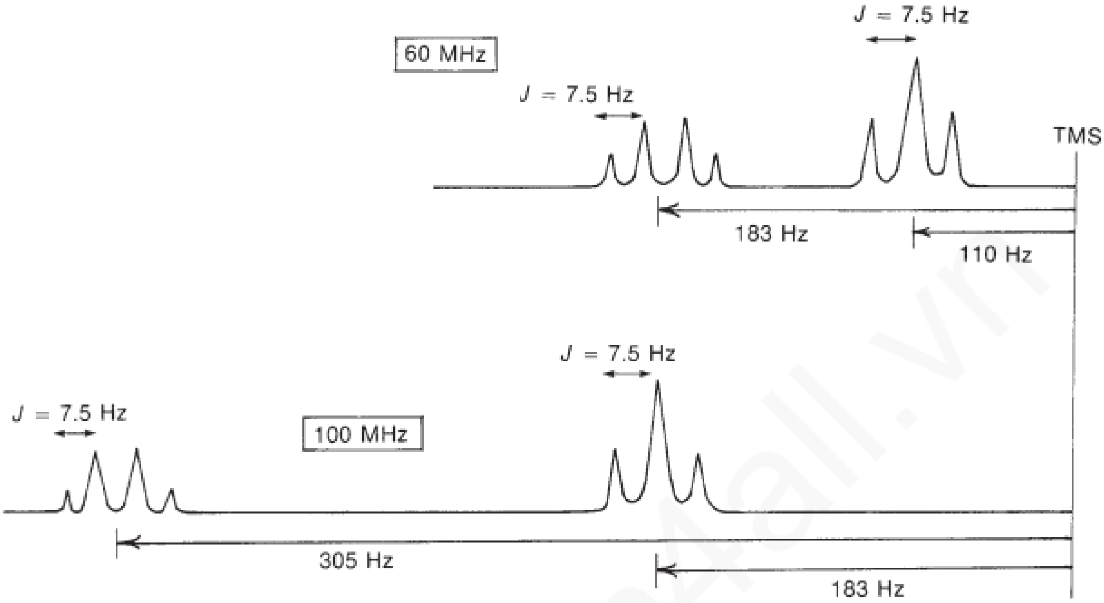
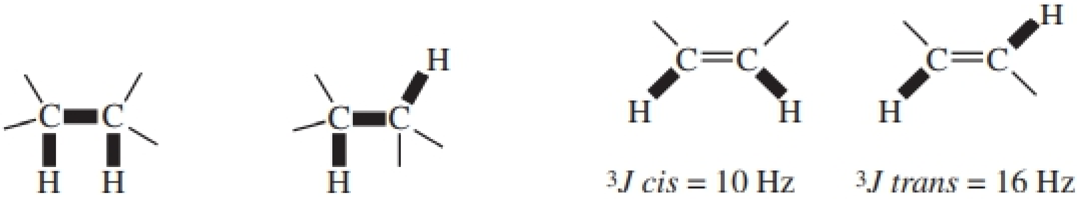


FIGURE 3.35 An illustration of the relationship between the chemical shift and the coupling constant.

Two hydrogen atoms on **adjacent carbon atoms** can be described as a three-bond interaction and abbreviated as ^3J. **Typical values for this most commonly observed coupling is approximately 6 to 8 Hz**. The bold lines in the diagram show how the hydrogen atoms are three bonds away from each other.

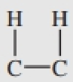
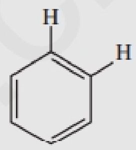
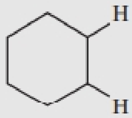
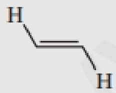
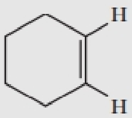
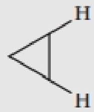
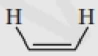
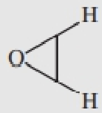
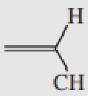
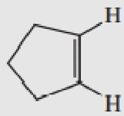


In alkenes, the ^3J coupling constants for hydrogen atoms that are **cis** to each other have values near **10 Hz**, while the ^3J coupling constants for hydrogen atoms that are **trans are larger, 16 Hz**.

A study of the magnitude of the coupling constant can give important structural information.

Table 3.9 gives the approximate values of some representative ^3J coupling constants. A more extensive list of coupling constants appears in Chapter 5, Section 5.2, and in Appendix 5. Before closing this section, we should take note of an axiom: the coupling constants of the groups of protons that split one another must be identical within experimental error.

TABLE 3.9
SOME REPRESENTATIVE ^3J COUPLING CONSTANTS AND THEIR APPROXIMATE VALUES (HZ)

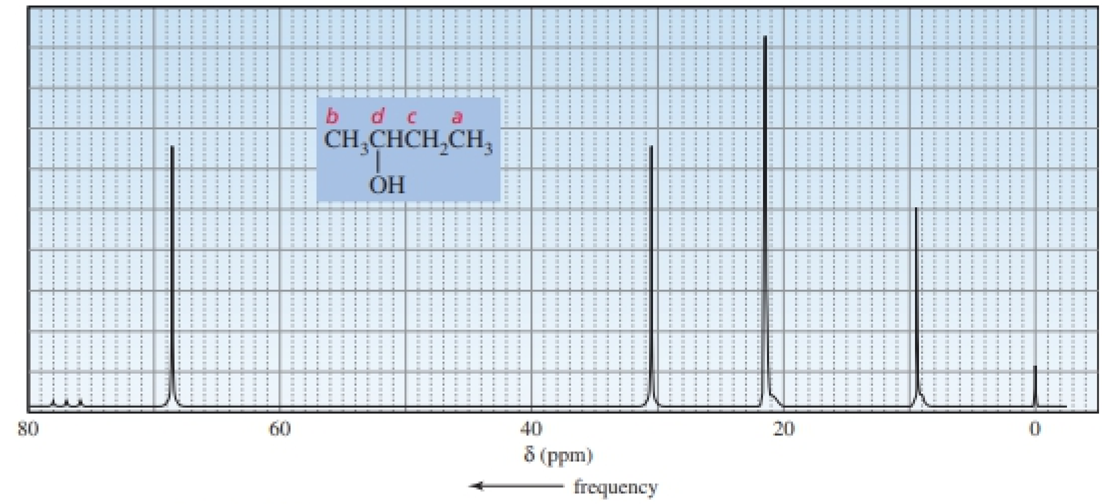
	6 to 8		*ortho* 6 to 10		a,a 8 to 14 a,e 0 to 7 e,e 0 to 5
	11 to 18		8 to 11		*cis* 6 to 12 *trans* 4 to 8
	6 to 15				*cis* 2 to 5 *trans* 1 to 3
	4 to 10				5 to 7

The number of signals in a NMR spectrum tells you how many different kinds of carbons a compound has—just as the number of signals in an NMR spectrum tells you how many different kinds of hydrogens a compound has. The principles behind 1 H 13C 13C ¢n J = 48 7 = 6.9 ¢n J = 240 7 = 34 Jac Ha Hc BRUI14-526_592r3 27-03-2003 3:10 PM Page 562 Section 14.18 NMR Spectroscopy 563 13C NMR and NMR spectroscopy are essentially the same. There are, however, some differences that make NMR easier to interpret. The development of NMR spectroscopy as a routine analytical procedure was not possible until computers were available that could carry out a Fourier transform (Section 14.2). NMR requires Fourier transform techniques because the signals obtained from a single scan are too weak to be distinguished from background electronic noise. However, FT– NMR scans can be repeated rapidly, so a large number of scans can be recorded and added. signals stand out when hundreds of scans are added, because electronic noise is random, so its sum is close to zero. Without Fourier transform, it could take days to record the number of scans required for a NMR spectrum. The individual signals are weak because the isotope of carbon that gives rise to NMR signals constitutes only 1.11% of carbon atoms (Section 13.3). (The most abundant isotope of carbon, has no nuclear spin and therefore cannot produce an NMR signal.) The low abundance of means that the intensities of the signals in NMR compared with those in NMR are reduced by a factor of approximately 100. In addition, the gyromagnetic ratio ( ) of is about one-fourth that of and the intensity of a signal is proportional to Therefore, the overall intensity of a signal is about 6400 times less than the intensity of an signal. One advantage to NMR spectroscopy is that the chemical shifts range over about 220 ppm, compared with about 12 ppm for NMR (Table 14.1). This means that signals are less likely to overlap. The NMR chemical shifts of different kinds of carbons are shown in Table 14.4. The reference compound used in NMR is TMS, the reference compound also used in NMR. Notice that ketone and aldehyde carbonyl groups can be easily distinguised from other carbonyl groups. A disadvantage of NMR spectroscopy is that, unless special techniques are used, the area under a NMR signal is not proportional to the number of atoms giving rise to the signal. Thus, the number of carbons giving rise to a NMR signal cannot routinely be determined by integration. The NMR spectrum of 2-butanol is shown in Figure 14.32. 2-Butanol has carbons in four different environments, so there are four signals in the spectrum. The relative positions of the signals depend on the same factors that determine the relative positions of the proton signals in NMR. Carbons in electron-dense environments produce low-frequency signals, and carbons close to electron-withdrawing groups produce high-frequency signals. This means that the signals for the carbons of 2-butanol are in the same relative order that we would expect for the signals of the protons on those carbons in the NMR spectrum. Thus, the carbon of the methyl group farther away from the electron-withdrawing OH group gives the lowest-frequency signal. As the frequency increases, the other methyl carbon comes next, followed by the methylene carbon; and the carbon attached to the OH group gives the highest-frequency signal.

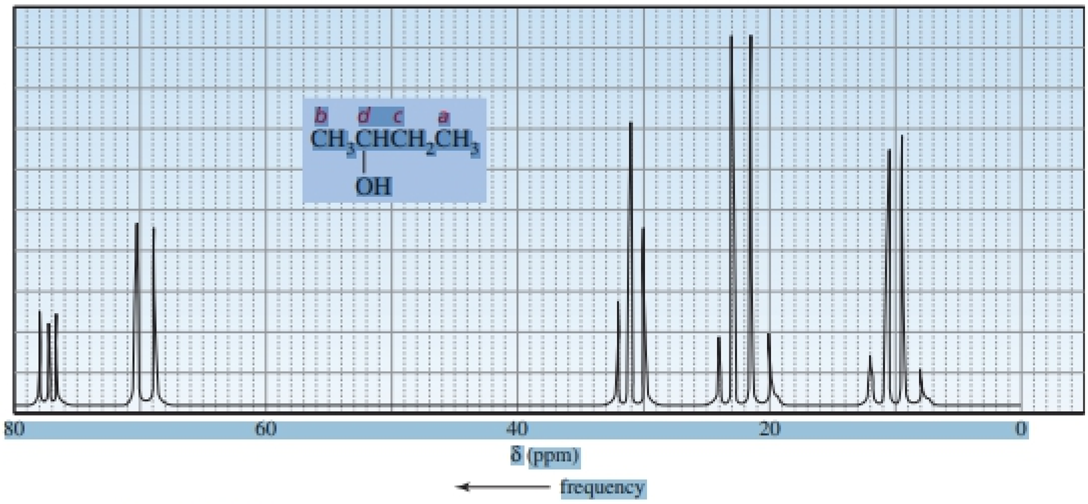
Table 14.4 Approximate Values of Chemical Shifts for ^{13}C NMR

Type of carbon	Approximate chemical shift (ppm)	Type of carbon	Approximate chemical shift (ppm)
$(CH_3)_4Si$	0	C—I	0–40
R—CH_3	8–35	C—Br	25–65
R—CH_2—R	15–50	C—Cl C—N C—O	35–80 40–60 50–80
R—CH(R)—R	20–60	R(—N)C=O	165–175
R—C(R)(R)—R	30–40	R(RO)C=O	165–175
≡C	65–85	R(HO)C=O	175–185
=C	100–150	R(H)C=O	190–200
 C	110–170	R(R)C=O	205–220

Proton-decoupled spectrum of 2-butanol. 13C NMR b dc a CH3CHCH2CH3 OH 80 60 40 20 0 δ (ppm) frequency ▲ Figure 14.33 Proton-coupled spectrum of 2-butanol. If the spectrometer is run in a proton-coupled mode, splitting is observed in a spectrum. 13C NMR 13C NMR The signals are not normally split by neighboring carbons because there is little likelihood of an adjacent carbon being a The probability of two carbons being next to each other is (about 1 in 10,000). (Because does not have a magnetic moment, it cannot split the signal of an adjacent)



▲ **Figure 14.32**



▲ **Figure 14.33**

NUCLEAR MAGNETIC RESONANCE SPECTROSCOPY Part Two: Carbon-13 Spectra, Including Heteronuclear Coupling with Other Nuclei The study of carbon nuclei through nuclear magnetic resonance (NMR) spectroscopy is an important technique for determining the structures of organic molecules. Using it together with proton NMR and infrared spectroscopy, organic chemists can often determine the complete structure of an unknown compound without "getting their hands dirty" in the laboratory! Fourier transform–NMR (FT-NMR) instrumentation makes it possible to obtain routine carbon spectra easily. Carbon spectra can be used to determine the number of nonequivalent carbons and to identify the types of carbon atoms (methyl, methylene, aromatic, carbonyl, and so on) that may be present in a compound. Thus, carbon NMR provides direct information about the carbon skeleton of a molecule. Some of the principles of proton NMR apply to the study of carbon NMR; however, structural determination is generally easier with carbon-13 NMR spectra than with proton NMR. Typically, both techniques are used together to determine the structure of an unknown compound. 177 CHAPTER 4 4.1 THE CARBON-13 NUCLEUS Carbon-12, the most abundant isotope of carbon, is NMR inactive since it has a spin of zero (see Section 3.1). Carbon-13, or 13C, however, has odd mass and does have nuclear spin, with I = –1 2 –. Unfortunately, the resonances of 13C nuclei are more difficult to observe than those of protons (1 H). They are about 6000 times weaker than proton resonances, for two major reasons. First, the natural abundance of carbon-13 is very low; only 1.08% of all carbon atoms in nature are 13C atoms. If the total number of carbons in a molecule is low, it is very likely that a majority of the molecules in a sample will have no 13C nuclei at all. In molecules containing a 13C isotope, it is unlikely that a second atom in the same molecule will be a 13C atom. Therefore, when we observe a 13C spectrum, we are observing a spectrum built up from a collection of molecules, in which each molecule supplies no more than a single 13C resonance. No single molecule supplies a complete spectrum. Second, since the magnetogyric ratio of a 13C nucleus is smaller than that of hydrogen (Table 3.2), 13C nuclei always have resonance at a frequency lower than protons. Recall that at lower frequencies, the excess spin population of nuclei is reduced (Table 3.3); this, in turn, reduces the sensitivity of NMR detection procedures. For a given magnetic field strength, the resonance frequency of a 13C nucleus is about one-fourth the frequency required to observe proton resonances (see Table 3.2). For example, in a 7.05-Tesla applied magnetic field, protons are observed at 300 MHz, while 13C nuclei are observed at about 75 MHz. With modern instrumentation, it is a simple matter to switch the transmitter frequency from the value required to observe proton resonances to the value required for 13C resonances.